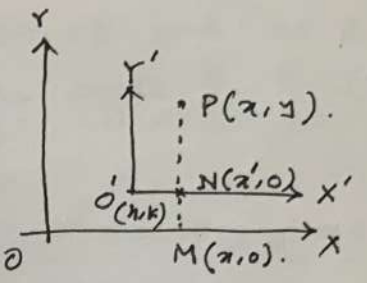
Transformation of co-ordinates.

Translation:

origin O, P is a point (x, y).

new " O' is at (h, k).

P w.r.t to O' be (x', y').



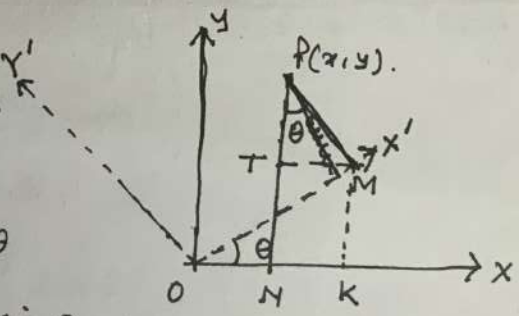
Then $x' = x - h$. & $y' = y - k$.

or, $x = x' + h$ & $y = y' + k$.

Thus $f(x, y) = 0$ after a translation of the origin to $O'(h, k)$ becomes $f(x' + h, y' + k) = 0$.

Rotation: Let θ be the angle through which the ~~axes~~ axes OX and OY, be turned to OX' and OY'.

Let (x, y) & (x', y') be the co-ord of P w.r.t. θ the old & new syst. respectively.



$OM = x'$, $ON = x$, $OK = x'\cos\theta$

$PM = y'$, $PN = y$, $TM = y'\sin\theta$

$PT = y'\cos\theta$, $MK = x'\sin\theta$

Now, $ON = x = OK - NK = x'\cos\theta - TM = x'\cos\theta - y'\sin\theta$.

i.e $x = x'\cos\theta - y'\sin\theta$. ——— (1)

Again, $PN = y = PT + TN = y'\cos\theta + MK$

$\Rightarrow y = y'\cos\theta + x'\sin\theta$ ——— (2)

Also, $x\cos\theta = x'\cos^2\theta - y'\sin\theta\cos\theta$ and

$y\sin\theta = y'\cos\theta\sin\theta + x'\sin^2\theta$.

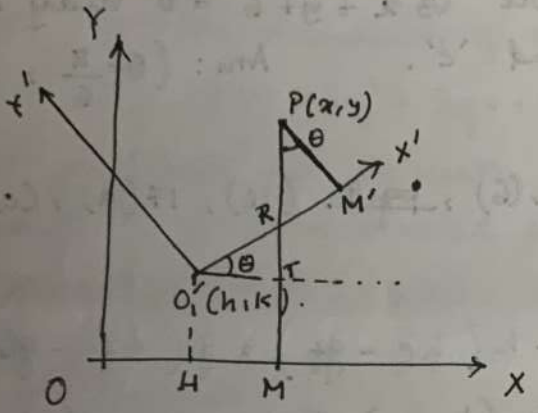
$\therefore x\cos\theta + y\sin\theta = x'$ Similarly $y\cos\theta - x\sin\theta = y'$

Note: Rotation is also known as orthogonal transformation

Translation + Rotation : Let $P(x, y)$ be a point w.r.t to O. Let O be shifted to $O'(h, k)$ and the axes OX and OY be turned through an angle θ. If (x', y') be the new co-or. w.r.t. O' then

$$x = h + x'\cos\theta - y'\sin\theta \quad , \quad y = k + x'\sin\theta + y'\cos\theta.$$

Proof



$OM = x \quad O'M' = x'$

$PM = y \quad PM' = y'$

$OH = h.$

In $\Delta PRM'$: $\tan\theta = \frac{RM'}{y'}$.

$\Rightarrow RM' = y'\tan\theta$

$OM = x = OH + HM \Rightarrow x = h + O'T$

Now, $O'T = O'R\cos\theta = (O'M' - RM')\cos\theta = x'\cos\theta - RM'\cos\theta$

$\Rightarrow O'T = x'\cos\theta - y'\tan\theta\cos\theta$

$= x'\cos\theta - y'\sin\theta.$

$\therefore x = h + x'\cos\theta - y'\sin\theta.$

Similarly $y = k + x'\sin\theta + y'\cos\theta.$

Problems.

1. To what point the origin is to be moved so that one can get rid of the first degree terms from the equation $x^2 + xy + 2y^2 - 7x - 5y + 12 = 0$. Ans: $\left(\frac{23}{7}, \frac{3}{7}\right)$.

2. Find the angle through which the axes are to be rotated so that the equation $\sqrt{3}x + y + 6 = 0$ may be reduced to the form $x = c$. Find 'c'. Ans: $\left(\theta = \frac{\pi}{6}, c = -3\right)$.

3. Examples 1. pg. 7 (3), (6), pg 9. (16), 17(a), (c), 19.

Sol^n 19. $\Delta = a(bc - f^2) - h(hc - gf) + g(hf - gb)$. —(1)

If the origin to shifted to (p, q) then, $x = x' + p$, $y = y' + q$. and the expression changes to

$$a(x'+p)^2 + 2h(x'+p)(y'+q) + b(y'+q)^2 + 2g(x'+p) + 2f(y'+q) + c. \quad —(2)$$

If the ~~exp~~ in the new co-ord. system is

$$a'x'^2 + 2h'x'y' + b'y'^2 + 2g'x' + 2f'y' + c'. \quad —(3)$$

Equating the co-efficients we have

$$\left.\begin{aligned} a &= a' \quad h = h' \quad b = b' \\ 2g' &= 2ap + 2hq + 2g \\ 2f' &= 2hp + 2bq + 2f. \\ c' &= ap^2 + 2hpq + bq^2 + 2gp + 2fq + c. \end{aligned}\right\} —(4).$$

The det. of (3) is

$$\Delta' = a'(b'c' - f'^2) - h'(h'c' - g'f') + g'(h'f' - g'b').$$

Replace $a', b', c', f', h', c', g'$ from (4) to get (1).

Center of a conic.

The point, any chord through which is bisected at that point is called the center of the conic.

Theorem : Equation of conic having its center at the origin is $ax^2 + 2hxy + by^2 + c = 0$.

Proof. Let the equation of the central conic be

$$ax^2 + 2hxy + by^2 + 2gx + 2fy + c = 0 \quad \text{- - - - (1)}$$

Let the conic has its center at the origin O. Then any chord through O will be bisected at O.

Let POP', $P \equiv (x_1, y_1)$ $P' \equiv (-x_1, -y_1)$ be any such chord. P & P' must satisfy (1).

$$\therefore \quad ax_1^2 + 2hx_1y_1 + by_1^2 + 2gx_1 + 2fy_1 + c = 0$$

$$ax_1^2 + 2hx_1y_1 + by_1^2 - 2gx_1 - 2fy_1 + c = 0.$$

subtracting we have

$$4gx_1 + 4fy_1 = 0 \Rightarrow gx_1 + fy_1 = 0$$

$\Rightarrow g = f = 0$, since (x_1, y_1) is a point other than origin.

Thus eq (1) becomes $ax^2 + 2hxy + by^2 + c = 0$.

Note : The first degree terms of the general equation of 2nd degree (1) vanishes if the origin be shifted to its centre.

This information is used in determining the center of the conic.

Determination of the centre of a conic.

Let (x_1, y_1) be the centre of the conic represented by

$$ax^2 + 2hxy + by^2 + 2gx + 2fy + c = 0. \quad \text{—(1)}$$

Shifting the origin to the centre we have (1) as

$$a(x'+x_1)^2 + 2h(x'+x_1)(y'+y_1) + b(y'+y_1)^2 + 2g(x'+x_1) + 2f(y'+y_1) + c = 0.$$

$$\Rightarrow ax'^2 + 2hx'y' + by'^2 + 2ax'x_1 + 2hx'y_1 + 2hx_1y' + 2hx_1y_1 + 2by'y_1 + by_1^2 + ax_1^2 + 2g(x'+x_1) + 2f(y'+y_1) + c = 0$$

$$\Rightarrow ax'^2 + 2hx'y' + by'^2 + 2(ax_1 + hy_1 + g)x' + 2(hx_1 + by_1 + f)y' + ax_1^2 + by_1^2 + 2hx_1y_1 + 2gx_1 + 2fy_1 + c = 0.$$

Since 1st degree terms must not be there

$\therefore ax_1 + hy_1 + g = 0$ and $hx_1 + by_1 + f = 0$. ----(2)

Solving (2) we have

$$x_1 = \frac{hf - bg}{ab - h^2}, \quad y_1 = \frac{gh - af}{ab - h^2}$$

Thus the centre is $\left(\frac{hf - bg}{ab - h^2}, \frac{gh - af}{ab - h^2}\right)$

1. For a conic (1) to have a centre, $ab - h^2 \neq 0$.

2. For a conic (1) having no centre, $ab - h^2 = 0$, and $hf - bg \neq 0$, $gh - af \neq 0$

3. For a conic (1) to have infinitely many centers $ab - h^2 = 0$, $hf - bg = 0$, $gh - af = 0$

General Equation of 2nd degree.

$$ax^2 + 2hxy + by^2 + 2gx + 2fy + c = 0 \quad \ldots\ldots (1)$$

let $\Delta = \begin{vmatrix} a & h & g \\ h & b & f \\ g & f & c \end{vmatrix}$ $\quad D = ab - h^2$

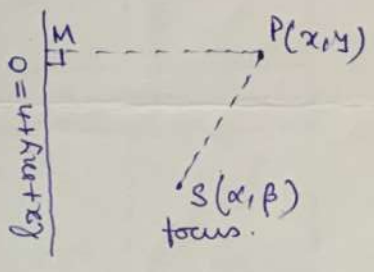
1. If $\Delta = 0$ then (1) represents a pair of straight lines.
   If $D = 0$ then the straight lines are parallel or co-incident.
   If $D \neq 0$ then the straight lines are intersecting.

2. If $\Delta \neq 0$ then (1) represents a proper conic.
   i.e a parabola, ellipse or hyperbola.

For a proper conic

$$PS^2 = e^2 PM^2$$

$$\Rightarrow (x-\alpha)^2 + (y-\beta)^2 = e^2\left[\frac{(lx+my+n)^2}{l^2+m^2}\right]$$



$$\Rightarrow x^2 + \alpha^2 - 2x\alpha + y^2 + \beta^2 - 2y\beta = \frac{e^2(l^2x^2 + m^2y^2 + n^2 + 2lmxy + 2mny + 2lnx)}{l^2+m^2}$$

$$\Rightarrow x^2(l^2+m^2-e^2l^2) + y^2(l^2+m^2-e^2m^2) - 2e^2xylm - 2x\{(l^2+m^2)\alpha + e^2nl\} - 2y\{\beta(l^2+m^2) + 2mne^2\} + (\alpha^2+\beta^2)(l^2+m^2) - n^2e^2 = 0. \quad \text{---(2)}$$

comparing (2) with (1) we have

$$a = l^2(1-e^2) + m^2 \qquad b = m^2(1-e^2) + l^2 \qquad h = -e^2lm$$

$$D = ab - h^2 = \{l^2(1-e^2)+m^2\}\{m^2(1-e^2)+l^2\} - l^2m^2e^4$$
$$= l^2m^2(1-e^2)^2 + l^4(1-e^2) + m^4(1-e^2) + m^2l^2 - l^2m^2e^4$$
$$= (1-e^2)\{l^2m^2(1-e^2) + l^4 + m^4 + l^2m^2(1+e^2)\}.$$

$\Rightarrow D = (1-e^2)(l^2+m^2)^2$

Since, for a parabola $e=1 \therefore D=0$

" an ellipse $e<1 \therefore D>0$

" a hyperbola $e>1 \therefore D<0$.

3. Let the axes be rotated through an acute angle θ. The co-ord. (x, y) transforms to (x', y').

$$x = x'\cos\theta - y'\sin\theta \quad ; \quad y = x'\sin\theta + y'\cos\theta.$$

Then (1) becomes

$$a(x'\cos\theta - y'\sin\theta)^2 + 2h(x'\cos\theta - y'\sin\theta)(x'\sin\theta + y'\cos\theta)$$
$$+ b(x'\sin\theta + y'\cos\theta)^2 + 2g(x'\cos\theta - y'\sin\theta)$$
$$+ 2f(x'\sin\theta + y'\cos\theta) + c = 0$$

$$\Rightarrow x'^2(a\cos^2\theta + 2h\cos\theta\sin\theta + b\sin^2\theta)$$
$$+ y'^2(a\sin^2\theta - 2h\cos\theta\sin\theta + b\cos^2\theta)$$
$$+ x'y'(-2a\cos\theta\sin\theta - 2h\sin^2\theta + 2h\cos^2\theta + 2b\sin\theta\cos\theta)$$
$$+ 2(g\cos\theta + f\sin\theta)x' + 2(-g\sin\theta + f\cos\theta)y' + c = 0.$$

For $x'y'$ term to vanish we must have

$$2(b-a)\sin\theta\cos\theta + 2h(\cos^2\theta - \sin^2\theta) = 0.$$

$$\Rightarrow (b-a)\sin 2\theta + 2h\cos 2\theta = 0 \Rightarrow \tan 2\theta = \frac{2h}{a-b}$$

$$\Rightarrow \theta = \frac{1}{2}\tan^{-1}\frac{2h}{a-b}.$$

Thus eqn (1) transforms to

$$Ax'^2 + By'^2 + 2Gx' + 2Fy' + c = 0. \quad \text{—— (3).}$$

Since Δ and D are invariant hence

$$\Delta = \begin{vmatrix} A & 0 & G \\ 0 & B & F \\ G & F & c \end{vmatrix} \qquad D = AB.$$

1. If $\Delta = 0$ then. ~~Eq (3) becomes~~

$$\begin{vmatrix} A & 0 & G \\ 0 & B & F \\ G & F & C \end{vmatrix} = A(BC - F^2) + G(-GB)$$

$$= ABC - AF^2 - BG^2$$

$$= AB\left[C - \frac{F^2}{B} - \frac{G^2}{A}\right]$$

$$= -K AB \quad \text{where } K = C - \frac{F^2}{B} - \frac{G^2}{A}$$

Eq^n (3) becomes

$$A\left(x' + \frac{G}{A}\right)^2 + B\left(y' + \frac{F}{B}\right)^2 = \frac{G^2}{A} + \frac{F^2}{B} - C$$

$$= K.$$

$$\Rightarrow \quad AX^2 + BY^2 = K \quad ; \quad x' + \frac{G}{A} = X \;,\; y' + \frac{F}{B} = Y.$$

If $D \neq 0$ then $A \neq 0$ and $B \neq 0$.

$\therefore$ $AX^2 + BY^2 = K$ represents a pair of intersecting st. lines.

If $D = 0$ then $A = 0$ or $B = 0$ or Both.

If $A = 0$ then, $BY^2 = K$.

If $B = 0$ then $AX^2 = K$

} In either case if $K = 0$ then $y^2 = 0$, $x^2 = 0$ represents a pair of co-incident st. lines and if $K \neq 0$ then $y^2 = K/B$ & $x^2 = K/A$ represents a pair of parallel st. lines.

2.

I. When $\Delta \neq 0$, $D = 0$

Since $D = AB = 0$ then $A = 0$ or $B = 0$ or both.

If $A = 0$ then (3) becomes

$$By'^2 + 2Gx' + 2Fy' + C = 0$$

$$\Rightarrow B\left(y' + \frac{F}{B}\right)^2 + 2Gx' + C = 0 + \frac{F^2}{B}$$

$$\Rightarrow B\left(y' + \frac{F}{B}\right)^2 = -2G\left(x' + \frac{C}{2G} - \frac{F^2}{2BG}\right).$$

or, $$\left(y' + \frac{F}{B}\right)^2 = -\frac{2G}{B}\left(x' + \frac{C}{2G} - \frac{F^2}{2BG}\right) \Rightarrow Y^2 = k.X$$

where $y' + \frac{F}{B} = Y$, $x' + \frac{C}{2G} - \frac{F^2}{2BG} = X$, $-\frac{2G}{B} = k$.

Similarly If $B = 0$ then, (3) reduces to

$X^2 = k'Y$ where $x' + \frac{G}{A} = X$, $y' + \frac{C}{2F} - \frac{G^2}{2FA}$, $-\frac{2F}{A} = k'$

If both $A = B = 0$ then $\Delta = \begin{vmatrix} A & 0 & G \\ 0 & B & F \\ G & F & C \end{vmatrix} = A(BC - F^2) + G(-BG)$

$$= ABC - AF^2 - BG^2 = 0$$

which is a contradiction.

Thus $A = 0$ or $B = 0$ (but not both) with $\Delta \neq 0$ represents a parabola.

2.

II. When $\Delta \neq 0$, $D > 0$

Then $D = AB > 0 \Rightarrow A \neq 0$, $B \neq 0$ and both of them are +ve or -ve.

Let us consider both A & B to be +ve.

Then, (3) may be written as

$$A\left(x' + \frac{G}{A}\right)^2 + B\left(y' + \frac{F}{B}\right)^2 - \frac{G^2}{A} - \frac{F^2}{B} + C = 0$$

$$\Rightarrow A\left(x' + \frac{G}{A}\right)^2 + B\left(y' + \frac{F}{B}\right)^2 = \frac{G^2}{A} + \frac{F^2}{B} - C$$

$$= K \text{ (says).}$$

$$\Rightarrow \frac{X^2}{K/A} + \frac{Y^2}{K/B} = 1 \quad \text{if} \quad x' + \frac{G}{A} = X \; ; \; y' + \frac{F}{B} = Y.$$

Since $\Delta = \begin{vmatrix} A & 0 & G \\ 0 & B & F \\ G & F & C \end{vmatrix} = A(BC - F^2) + G(-GB)$

$$= ABC - AF^2 - BG^2$$

$$= AB\left[C - \frac{F^2}{B} - \frac{G^2}{A}\right].$$

$$= -K AB.$$

(i) If $K > 0$ then $\Delta < 0$ and the ellipse is real.

(ii) If $K = 0$ then $\Delta = 0$ and the " " null.
not possible.

(iii) If $K < 0$ then $\Delta > 0$ and " " " imaginary

2.

III. $\Delta \neq 0$, $D < 0$.

Then $A \neq 0$, $B \neq 0$ and A and B are of opposite signs.

Assume $A > 0$ and $B < 0$.

Eq^n (3) becomes (as in II)

$$\frac{X^2}{K/A} + \frac{Y^2}{K/B} = 1.$$ since $B < 0$ $\therefore$ it is a hyperbola.

Here $\Delta = -ABK$.

If $K > 0$ or $K < 0$, both cases represent a hyperbola.

If $K = 0$ then $AX^2 + BY^2 = 0$ $(B < 0)$ represents a pair of straight lines which are assymptotes to the given hyperbola.

2. If $P(9,-1)$ and $Q(-3,4)$ are two points and the origin is shifted to P and the co-ordinates axes are rotated so that the positive direction of the new x-axis agrees with the direction of the segment $\overline{PQ}$ then find the co-ord. transformation formulae.

Sol: Slope of $PQ = \frac{4+1}{-12} = -\frac{5}{12}$ $\therefore \tan\theta = -\frac{5}{12}$

$\theta = \tan^{-1}\left(-\frac{5}{12}\right)$ ——— (1)

Thus the axes are rotated through an angle θ given by (1) If the origin is shifted to $P(9,-1)$ and (x', y') be the new co-ord. of a point then

$$x' = 9 + x\cos\theta - y\sin\theta$$

$$y' = -1 + x\sin\theta + y\cos\theta \quad ; \theta \text{ given by (1)}$$

3. If by a rotation of co-ord. axes, the expression $ax^2 + 2hxy + by^2 + 2gx + 2fy + c$ is changed into $a'x'^2 + 2h'x'y' + b'y'^2 + 2g'x' + 2f'y' + c'$; show that $b'c' + c'a' + a'b' - f'^2 - g'^2 - h'^2 = bc + ca + ab - f^2 - g^2 - h^2$.

Sol^n. Let the axes be turned through an angle θ.
The new co-ord. of a point (x, y) be (x', y').

Then, $x' = x\cos\theta + y\sin\theta$; $y' = -x\sin\theta + y\cos\theta$. — (1)
Or, $x = x'\cos\theta - y'\sin\theta$; $y = x'\sin\theta + y'\cos\theta$
The expression $ax^2 + 2hxy + by^2 + 2gx + 2fy + c$ becomes — (2)

$$a(x'\cos\theta - y'\sin\theta)^2 + 2h(x'\cos\theta - y'\sin\theta)(x'\sin\theta + y'\cos\theta) + b(x'\sin\theta + y'\cos\theta)^2 + 2g(x'\cos\theta - y'\sin\theta) + 2f(x'\sin\theta + y'\cos\theta) + c$$

$$= x'^2(a\cos^2\theta + 2h\sin\theta\cos\theta + b\sin^2\theta) + y'^2(a\sin^2\theta - 2h\sin\theta\cos\theta + b\cos^2\theta) + x'y'(-2a\cos\theta\sin\theta - 2h\sin^2\theta + 2h\cos^2\theta + 2b\sin\theta\cos\theta) + x'(2g\cos\theta + 2f\sin\theta) + y'(-2g\sin\theta + 2f\cos\theta) + c$$
— (3)

If (2) changes to $a'x'^2 + 2h'x'y' + b'y'^2 + 2g'x' + 2f'y' + c'$
Then from (3)

$$a' = a\cos^2\theta + h\sin 2\theta + b\sin^2\theta$$
$$b' = a\sin^2\theta - h\sin 2\theta + b\cos^2\theta$$
$$h' = \cos\theta\sin\theta(b - a) + h(\cos 2\theta).$$
$$g' = g\cos\theta + f\sin\theta$$
$$f' = -g\sin\theta + f\cos\theta. \quad \& \quad c' = c.$$

Now, $b'c' + a'c' + a'b' - f'^2 - g'^2 - h'^2$ — (4)

$$c'(a' + b') = c(a + b)$$
$$a'b' = a^2\cos^2\theta\sin^2\theta - ah\cos^2\theta\sin 2\theta + ab\cos^4\theta + b^2\sin^2\theta\cos^2\theta + ah\sin^2\theta\sin 2\theta + ab\sin^4\theta - h^2\sin^2 + hb\cos^2\theta\sin^2\theta - bh\sin^2\theta\sin 2\theta$$
$$= (a^2 + b^2)(\cos\theta\sin\theta)^2 - ah\cos 2\theta\sin 2\theta + hb\cos 2\theta$$

$$f'^2 + g'^2 = (-g\sin\theta + f\cos\theta)^2 + (g\cos\theta + f\sin\theta)^2$$

$$= g^2 + f^2$$

$\therefore$ Exp. (4) becomes

$$c(a+b) + a'b' - h'^2 - (g^2+f^2)$$

$$= c(a+b) + (a^2+b^2)(\cos\theta\sin\theta)^2 - h\cos 2\theta \sin 2\theta (a-b)$$
$$+ ab(\cos^4\theta + \sin^4\theta) - h^2\sin^2 2\theta$$
$$- h^2\cos^2 2\theta - h\cos 2\theta \sin 2\theta (b-a) - (b-a)^2(\cos\theta\sin\theta)^2$$
$$- (g^2+f^2)$$

$$= c(a+b) - g^2 - f^2 + \cos^2\theta\sin^2\theta\,[a^2+b^2-b^2-a^2+2ab]$$
$$+ ab(\cos^4\theta + \sin^4\theta) - h^2(\sin^2 2\theta + \cos^2 2\theta)$$

$$= c(a+b) - g^2 - f^2 + ab\,[\cos^4\theta + \sin^4\theta + 2\cos^2\theta\sin^2\theta]$$
$$- h^2$$

$$= bc + ac + ab - f^2 - g^2 - h^2 \quad \text{proved.}$$

1. Find the angle through which axes must be turned so that $ax^2 + 2hxy + by^2$ becomes an expression of the form $AX^2 + BY^2$.

Sol^n. Let the axes be turned through an angle θ and the new co-ord. of a point (x, y) be (X, Y). Then

$$x = X\cos\theta - Y\sin\theta \quad , \quad y = X\sin\theta + Y\cos\theta \qquad \text{—(1)}$$

The expression $ax^2 + 2hxy + by^2$ transforms to

$$a(X\cos\theta - Y\sin\theta)^2 + 2h(X\cos\theta - Y\sin\theta)(X\sin\theta + Y\cos\theta) + b(X\sin\theta + Y\cos\theta)^2 \qquad \text{—(2)}$$

$$= (a\cos^2\theta + 2h\cos\theta\sin\theta + b\sin^2\theta)X^2 + (a\sin^2\theta - 2h\sin\theta\cos\theta + b\cos^2\theta)Y^2 + 2(-a\cos\theta\sin\theta - h\sin^2\theta + h\cos^2\theta + b\sin\theta\cos\theta) \qquad \text{—(3)}$$

Since $AX^2 + BY^2$ contains no term with XY
$\therefore$ the co-eff. of XY in (3) must be zero.

i.e $(b-a)\cos\theta\sin\theta + h(\cos^2\theta - \sin^2\theta) = 0$.

$\Rightarrow (b-a)\sin 2\theta + 2h\cos 2\theta = 0$

$\Rightarrow \tan 2\theta = -\frac{2h}{b-a} = \frac{2h}{a-b}$.

$\Rightarrow \theta = \frac{1}{2}\tan^{-1}\left(\frac{2h}{a-b}\right)$

2. Reduce the equation $x^2 + 24xy - 6y^2 + 28x + 36y + 16 = 0$ to standard form and hence find the centre and eccentricity of the conic represented by it.

Determine the angle through which the axis must be turned so that the equation $lx+my+n=0$ $(m \neq 0)$ may reduce to the form $by+c=0$.

Sol^n: Let the angle through which the axes be turned be θ. and the new co-ord of a Point $P(x,y)$ w.r.to new axes be (x', y').

Then transformation rules are

$$\left.\begin{aligned} x &= x'\cos\theta - y'\sin\theta \\ y &= x'\sin\theta + y'\cos\theta \end{aligned}\right\} \quad -(1)$$

Then the eq^n $lx+my+n=0$ transforms to

$$l(x'\cos\theta - y'\sin\theta) + m(x'\sin\theta + y'\cos\theta) + n = 0$$

$$\Rightarrow \quad x'(l\cos\theta + m\sin\theta) + y'(-l\sin\theta + m\cos\theta) + n = 0.$$

By the given conditions

$$l\cos\theta + m\sin\theta = 0 \Rightarrow \quad m\sin\theta = -l\cos\theta$$

$$\Rightarrow \quad \tan\theta = -\frac{l}{m}.$$

$$\therefore \quad \theta = \tan^{-1}\left(-\frac{l}{m}\right) = -\tan^{-1}\left(\frac{l}{m}\right); \quad m \neq 0$$

2. Reduce the eq^n of the conic $x^2 - 2xy + y^2 + 6x - 14y + 29 = 0$ to canonical form and hence find the directrix and vertex of the conic.

Sol^n. $a=1 \quad b=1 \quad h=-1 \quad g=3 \quad f=-7, \quad c=29$

$$\Delta = \begin{vmatrix} a & h \\ h & b \end{vmatrix} = \begin{vmatrix} 1 & -1 \\ -1 & 1 \end{vmatrix} = 1-1 = 0$$

Thus it is a non central conic.

Let us apply a rotation through acute angle θ so tha

Thus $(x'\cos\theta - y'\sin\theta)^2 - 2(x'\cos\theta - y'\sin\theta)(x'\sin\theta + y'\cos\theta)$

$+ (x'\sin\theta + y'\cos\theta)^2 + 6(x'\cos\theta - y'\sin\theta)$

$-14(x'\sin\theta + y'\cos\theta) + 29 = 0$.

co-eff of $x'y'$ is $= 0$

$\Rightarrow -2\cos\theta\sin\theta - 2(\cos^2\theta - \sin^2\theta) + 2\cos\theta\sin\theta = 0$

$\Rightarrow -2\cos 2\theta = 0 \Rightarrow \cos 2\theta = 0$

$\Rightarrow$ to $2\theta = \frac{\pi}{2}$ ~~0 or π~~. $\Rightarrow \theta = \frac{\pi}{4}$

$\therefore \theta = \pi/4$ is admissible.

Thus eqn becomes on putting $\cos\frac{\pi}{4} = \frac{1}{\sqrt{2}}$ & $\sin\frac{\pi}{4} = \frac{1}{\sqrt{2}}$

~~$y'^2\sin^2\frac{\pi}{2} - 2(-y'\sin$~~

$+ y'^2 - 2(-y')(x') + x'^2 - 6y' - 14x' + 29 = 0$

$\Rightarrow y'^2 + 2x'y' + x'^2 - 6y' - 14x' + 29 = 0$.

$\Rightarrow \left(\frac{x'-y'}{\sqrt{2}}\right)^2 - 2\left(\frac{x'-y'}{\sqrt{2}}\right)\left(\frac{x'+y'}{\sqrt{2}}\right) + \left(\frac{x'+y'}{\sqrt{2}}\right)^2$

$+ 6\left(\frac{x'-y'}{\sqrt{2}}\right) - 14\left(\frac{x'+y'}{\sqrt{2}}\right) + 29 = 0$

$\Rightarrow \frac{1}{2}(x'^2 - 2x'y' + y'^2) - (x'^2 - y'^2)$

$+ \frac{1}{2}(x'^2 + 2x'y' + y'^2) + \left(\frac{6}{\sqrt{2}} - \frac{14}{\sqrt{2}}\right)x'$

$+ \left(-\frac{6}{\sqrt{2}} - \frac{14}{\sqrt{2}}\right)y' + 29 = 0$

$\Rightarrow x'^2 + y'^2 - x'^2 + y'^2 - \frac{8}{\sqrt{2}}x' - \frac{20}{\sqrt{2}}y' + 29 = 0$

$\Rightarrow 2y'^2 - \frac{8}{\sqrt{2}}x' - \frac{20}{\sqrt{2}}y' + 29 = 0$.

$$\Rightarrow y'^2 - \frac{4x'}{\sqrt{2}} - \frac{10y'}{\sqrt{2}} + \frac{29}{2} = 0$$

$$\Rightarrow \left(y' - \frac{5}{\sqrt{2}}\right)^2 - \frac{4x'}{\sqrt{2}} + \frac{29}{2} - \frac{25}{2} = 0$$

$$\Rightarrow \left(y' - \frac{5}{\sqrt{2}}\right)^2 - \frac{4x'}{\sqrt{2}} + \frac{4}{2} = 0 \Rightarrow \left(y' - \frac{5}{\sqrt{2}}\right)^2 = \frac{4}{\sqrt{2}}\left(x' - \frac{1}{\sqrt{2}}\right)$$

Let the origin be shifted to $\left(\frac{1}{\sqrt{2}}, \frac{5}{\sqrt{2}}\right)$.

Then $X = x' - 1/\sqrt{2}$ & $Y = y' - 5/\sqrt{2}$

and $Y^2 = 4 \cdot \frac{1}{\sqrt{2}} X$. is the eqn of a parabola.

whose vertex is $\left(\frac{1}{\sqrt{2}}, \frac{5}{\sqrt{2}}\right)$ and directrix is $X = \frac{1}{\sqrt{2}}$

i.e $x' - \frac{1}{\sqrt{2}} = \frac{1}{\sqrt{2}} \Rightarrow x' = \sqrt{2}$ and so on.

2. Determine whether the following conic has a single center, no center or infinitely many centers.

$$x^2 - 4xy + 4y^2 - 8x + 6y + 10 = 0.$$

Sol^n Let the given conic has a center at (α, β). Shifting the origin to the center (α, β) we have the new co-ord. of the point (x, y) as (x', y') given as

$$x = x' + \alpha \quad \text{and} \quad y = y' + \beta$$

The eq^n becomes

$$(x' + \alpha)^2 - 4(x' + \alpha)(y' + \beta) + 4(y' + \beta)^2 - 8(x' + \alpha) + 6(y' + \beta) + 10 = 0.$$

The co-efficients of the first degree terms must be zero.

$\therefore \quad 2\alpha - 4\beta - 8 = 0 \Rightarrow \alpha - 2\beta - 4 = 0$

and $-4\alpha + 8\beta + 6 = 0. \Rightarrow \alpha - 2\beta - \frac{6}{4} = 0.$

Solving for α and β we have

$$\frac{\alpha}{2 \times \frac{6}{4} - 8} = \frac{\beta}{-4 + \frac{6}{4}} = \frac{1}{-2 + 2}$$

$$\Rightarrow \alpha = -\frac{5}{0}, \quad \beta = -\frac{10/4}{0}$$

Thus the conic has no center.

3. Reducing the equation to its canonical form determine the nature of the conic for different values of a.

$$4x^2 + 4xy + y^2 - 4x - 2y + a = 0. \quad \text{——(1)}$$

Sol^n Here $a = 4 \quad h = 2 \quad b = 1 \quad g = -2 \quad f = -1 \quad c = a.$

$D = ab - h^2 = 4 - 2^2 = 0$ and

$$\Delta = \begin{vmatrix} a & h & g \\ h & b & f \\ g & f & c \end{vmatrix} = \begin{vmatrix} 4 & 2 & -2 \\ 2 & 1 & -1 \\ -2 & -1 & a \end{vmatrix}$$

$$= 4(a-1) - 2(2a-2) - 2(-2+2)$$
$$= 4a - 4 - 4a + 4 = 0$$

Thus (1) represents a pair of non intersecting straight lines.

Thus $4x^2 + 4xy + y^2 - 4x - 2y + a = 0$

$$\Rightarrow 4x^2 + (4y-4)x + y^2 - 2y + a = 0$$

$$\Rightarrow x = \frac{-4(y-1) \pm \sqrt{16(y-1)^2 - 16(y^2-2y+a)}}{2 \times 4}$$

$$= \frac{1}{2}\left\{(1-y) \pm \sqrt{1-a}\right\}$$

$$\Rightarrow 2x = 1 - y \pm \sqrt{1-a} \quad \Rightarrow 2x + y = 1 \pm \sqrt{1-a}.$$

If $a = 1$ then $2x + y = 1$ represents a single st. line.
Hence (1) represents a pair of co-incident st. lines.

If $a < 1$ then $(1-a) > 0$. Hence (1) → pair of $\parallel$ st. lines
If $a > 1$ then $(1-a) < 0$. " " → imaginary st. lines

Ex. 1. *Reduce the following equations to their canonical forms and determine the nature of the conics represented by them :*

(*a*) $3x^2+2xy+3y^2-16x+20=0$;

(*b*) $x^2+4xy+y^2-2x+2y+6=0$;

(*c*) $6x^2-5xy-6y^2+14x+5y+4=0$;

(*d*) $4x^2+4xy+y^2-12x-6y+5=0$. [*C. H. 1981*]

(*a*) Here $\Delta = 3.3.20+2.0.(-8).1-3.0-3.(-8)^2-20(1)^2$

$= -32 \neq 0$

and $D = 3.3-1^2 = 8 \neq 0$.

Hence it is a central conic.

Let the centre of the conic be at (α, β).

Transferring the origin to the point (α, β) without changing the directions of the axes, the equation becomes with (x', y') as new co-ordinates

$$3(x'+\alpha)^2+2(x'+\alpha)(y'+\beta)+3(y'+\beta)^2-16(x'+\alpha)+20=0$$

or, $3x'^2+2x'y'+3y'^2+2x'(3\alpha+\beta-8)+2y'(\alpha+3\beta)$

$$+(3\alpha^2+2\alpha\beta+3\beta^2-16\alpha+20)=0. \quad \cdots(1)$$

The origin being at the centre, coefficients of x' and y' should separately vanish, that is,

$3\alpha+\beta-8=0$ and $\alpha+3\beta=0$, whence $\alpha=3$, $\beta=-1$.

Thus the centre of the conic is at $(3, -1)$.

Then (1) becomes $3x'^2+2x'y'+3y'^2-4=0$.

To remove the $x'y'$-term, we rotate the axes through an angle θ.

Due to this rotation, the equation reduces to [with new co-ordinates (X, Y)]

$$3(X\cos\theta - Y\sin\theta)^2+2(X\cos\theta - Y\sin\theta)(X\sin\theta + Y\cos\theta)$$

$$+3(X\sin\theta + Y\cos\theta)^2=4$$

or, $X^2(3+\sin 2\theta)+2XY\cos 2\theta+Y^2(3-\sin 2\theta)=4. \quad \cdots (2)$

For the removal of XY-term, we require that 2θ which makes

$$\cos 2\theta = 0, \text{ i.e., } \theta = \tfrac{1}{4}\pi.$$

Then (2) becomes

$$X^2(3+1)+Y^2(3-1)=4$$

or, $\dfrac{X^2}{1}+\dfrac{Y^2}{2}=1.$

This is the required reduced canonical form of the given equation.

This represents an ellipse.

(*b*) Here $\Delta = 1.1.6 + 2.1.(-1)\,2 - 1.1^2 - 1\,(-1)^2 - 6.2^2$

$= -24 \neq 0$

and $D = 1 - 4 = -3 \neq 0.$

Hence it is a central conic.

Let the centre of the conic be at (α, β).

The equations determining the centre are

$\alpha + 2\beta - 1 = 0$ and $2\alpha + \beta + 1 = 0$, giving $\alpha = -1, \beta = 1.$

The reduced equation with the centre at the origin is

$$x'^2 + 4x'y' + y'^2 + \frac{\Delta}{D} = 0, \text{ i.e., } x'^2 + 4x'y' + y'^2 + 8 = 0.$$

If the finally reduced equation after rotation be

$$a'X^2 + b'Y^2 + 8 = 0,$$

then, by the theory of invariants,

$a' + b' = 1 + 1 = 2$ and $a'b' = 1 - 4 = -3.$

$\therefore$ $a' = -1, 3$ and corresponding $b' = 3, -1.$

Hence the reduced standard form of the given equation is

either $-X^2 + 3Y^2 + 8 = 0$. *i.e.*, $\dfrac{X^2}{8} - \dfrac{Y^2}{8/3} = 1$

or, $3X^2 - Y^2 + 8 = 0$, *i.e.*, $\dfrac{Y^2}{8} - \dfrac{X^2}{8/3} = 1$

each of which is a hyperbola.

Note. The transformation can also be effected by the rotation of axes through an angle θ, given by

$$\tan 2\theta = \frac{2h}{a - b}, \text{ giving } \theta = \frac{\pi}{4} \text{ in this case.}$$

(*c*) Here

$\Delta = 6.(-6).4 + 2.\frac{5}{2}.7.(-\frac{5}{2}) - 6.(\frac{5}{2})^2 - (-6)(7)^2 - 4.(-\frac{5}{2})^2 = 0$

and $D = -36 - \frac{25}{4} = -\frac{169}{4} \neq 0.$

Thus the given equation represents a pair of intersecting straight lines.

The equations giving their point of intersection are

$12\alpha - 5\beta + 14 = 0$, $-5\alpha - 12\beta + 5 = 0$, whence $\alpha = -\frac{11}{13}, \beta = \frac{10}{13}.$

Shifting the origin to the point (α, β), *i.e.*, to the point $(-\frac{11}{13}, \frac{10}{13})$, without changing the directions of the axes, the given equation reduces to $6x'^2 - 5x'y' - 6y'^2 = 0.$

Rotating the axes through an angle θ with the same origin, this equation becomes [with new co-ordinates (X, Y)]

$$6(X\cos\theta - Y\sin\theta)^2 - 5(X\cos\theta - Y\sin\theta)(X\sin\theta + Y\cos\theta) - 6(X\sin\theta + Y\cos\theta)^2 = 0$$

or, $(6\cos 2\theta - \frac{5}{2}\sin 2\theta)X^2 - (5\cos 2\theta + 12\sin 2\theta)XY - (6\cos 2\theta - \frac{5}{2}\sin 2\theta)Y^2 = 0.$

To remove the XY-term, we put $5\cos 2\theta + 12\sin 2\theta = 0$

or, $$\frac{\sin 2\theta}{-5} = \frac{\cos 2\theta}{12} = \frac{1}{13}.$$

With these values, the given equation reduces to $X^2 - Y^2 = 0$ which represents two intersecting straight lines

$$X - Y = 0,\ X + Y = 0.$$

(*d*) Here $\Delta = 4.1.5 + 2.(-3)(-6)2 - 4.(-3)^2 - 1.(-6)^2 - 5.2^2 = 0$

and $D = 4.1 - 2^2 = 0.$

Hence the given equation represents a pair of parallel straight lines.

Rotating the axes through an angle θ with the same origin, the given equation becomes [with new co-ordinates (X, Y)]

$$4(X\cos\theta - Y\sin\theta)^2 + 4(X\cos\theta - Y\sin\theta)(X\sin\theta + Y\cos\theta) + (X\sin\theta + Y\cos\theta)^2 - 12(X\cos\theta - Y\sin\theta) - 6(X\sin\theta + Y\cos\theta) + 5 = 0$$

or, $(\frac{5}{2} + 2\sin 2\theta + \frac{3}{2}\cos 2\theta)X^2 + (4\cos 2\theta - 3\sin 2\theta)XY + (\frac{5}{2} - 2\sin 2\theta - \frac{3}{2}\cos 2\theta)Y^2 - 2(6\cos\theta + 3\sin\theta)X + 2(6\sin\theta - 3\cos\theta)Y + 5 = 0.$

To remove the XY-term, we put $4\cos 2\theta - 3\sin 2\theta = 0$

or, $$\frac{\sin 2\theta}{4} = \frac{\cos 2\theta}{3} = \frac{1}{5}.$$

$\therefore$ $\sin\theta = \frac{1}{\sqrt{5}}$, $\cos\theta = \frac{2}{\sqrt{5}}$, taking the positive sign.

With these values, the given equation reduces to

$$5X^2 - 6\sqrt{5}X + 5 = 0$$

or, $$(X - \sqrt{5})\left(X - \frac{1}{\sqrt{5}}\right) = 0$$

which represents two parallel straight lines

$$X - \sqrt{5} = 0,\ X - \frac{1}{\sqrt{5}} = 0.$$

TANGENTS AND NORMALS

5·1. Tangents.

Let two points P and Q be taken on any curve and let the point Q move along the curve nearer and nearer to the point P; then the limiting position of the straight line PQ, when Q moves upto and ultimately coincides with P, is called the *tangent* to the curve at the point. More precisely, the tangent to a conic at a point is the limiting position of a straight line when the two points of intersection of the line and the conic coincide at the point.

5·2. Equation of the tangent to the circle $x^2+y^2+2gx+2fy+c=0$ at the point (x_1, y_1).

Let $P(x_1, y_1)$ be a point on the circle

$$x^2+y^2+2gx+2fy+c=0. \quad \cdots \quad (1)$$

Let Q be another point (x_2, y_2) on the circle very near to P. The equation of the straight line PQ is

$$y-y_1=\frac{y_2-y_1}{x_2-x_1}(x-x_1). \quad \cdots \quad (2)$$

Since both P and Q lie on the circle, we have

$$x_1^2+y_1^2+2gx_1+2fy_1+c=0, \quad \cdots \quad (3)$$

$$x_2^2+y_2^2+2gx_2+2fy_2+c=0. \quad \cdots \quad (4)$$

Subtracting (3) from (4), we get

$$x_2^2-x_1^2+y_2^2-y_1^2+2g(x_2-x_1)+2f(y_2-y_1)=0$$

or, $(x_2-x_1)(x_2+x_1+2g)+(y_2-y_1)(y_2+y_1+2f)=0.$

$$\therefore \quad \frac{y_2-y_1}{x_2-x_1}=-\frac{x_1+x_2+2g}{y_1+y_2+2f}.$$

Substituting this in (2), the equation becomes

$$y-y_1=-\frac{x_1+x_2+2g}{y_1+y_2+2f}(x-x_1).$$

Hence the equation of the chord PQ is

$$(y-y_1)(y_1+y_2+2f)+(x-x_1)(x_1+x_2+2g)=0. \quad \cdots \quad (5)$$

As Q approaches and coincides with P, the chord PQ becomes the tangent at P.

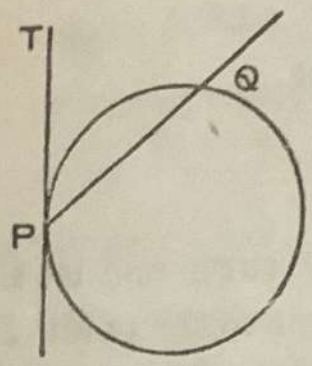


Fig. 19

Substituting x_1 for x_2 and y_1 for y_2 in (5), we get the equation of the tangent at $P(x_1, y_1)$ as

$$(y-y_1)(2y_1+2f)+(x-x_1)(2x_1+2g)=0$$

or, $x(x_1+g)+y(y_1+f)=x_1^2+y_1^2+gx_1+fy_1$.

Adding (gx_1+fy_1+c) to both sides and using (3), we get the equation of the tangent as

$$xx_1+yy_1+g(x+x_1)+f(y+y_1)+c=0.$$

5·3. Equation of the tangent to the parabola $y^2 = 4ax$ at the point (x_1, y_1).

Let $P(x_1, y_1)$ and $Q(x_2, y_2)$ be two very close points on the parabola

$$y^2 = 4ax. \quad \cdots \quad (1)$$

The equation of the straight line PQ is

$$y - y_1 = \frac{y_2 - y_1}{x_2 - x_1}(x - x_1). \quad \cdots \quad (2)$$

The points $P(x_1, y_1)$ and $Q(x_2, y_2)$ lie on the parabola.

Hence $y_1^2 = 4ax_1$ and $y_2^2 = 4ax_2$, $\cdots$ (3)

$\therefore \quad y_2^2 - y_1^2 = 4a(x_2 - x_1)$

$$i.e., \quad \frac{y_2 - y_1}{x_2 - x_1} = \frac{4a}{y_2 + y_1}.$$

Substituting in (2), the equation of the chord PQ is

$$y - y_1 = \frac{4a}{y_2 + y_1}(x - x_1).$$

As Q approaches and coincides with P, the chord PQ becomes the tangent at P.

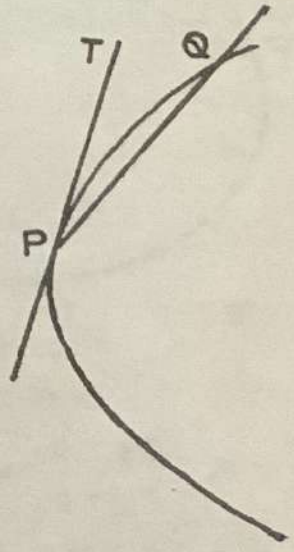


Fig. 20

Substituting x_1 for x_2 and y_1 for y_2, the equation of the tangent becomes

$$y - y_1 = \frac{4a}{2y_1}(x - x_1)$$

or, $\quad (y - y_1)y_1 = 2a(x - x_1)$

or, $\quad yy_1 - 2ax = y_1^2 - 2ax_1.$

Adding $(-2ax_1)$ to both sides,

$$yy_1 - 2a(x+x_1) = y_1^2 - 4ax_1 = 0, \text{ by (3).}$$

$\therefore$ The equation of the tangent at $P(x_1, y_1)$ is

$$yy_1 = 2a(x+x_1).$$

Cor. The tangent at (0, 0) to the parabola $y^2 = 4ax$ is $x = 0$, that is, the tangent at the vertex is perpendicular to the axis of the parabola. The equation of the tangent to the parabola $y^2 = 4ax$ at the point $(at^2, 2at)$ is

$$x - ty + at^2 = 0.$$

Note. The equation of the tangent may also be obtained by the second method of Art. 5·2.

5·4. Equation of the tangent to the ellipse $\frac{x^2}{a^2} + \frac{y^2}{b^2} = 1$ at the point (x_1, y_1).

Let $P(x_1, y_1)$ and $Q(x_2, y_2)$ be two very close points on the ellipse

$$\frac{x^2}{a^2} + \frac{y^2}{b^2} = 1. \quad \ldots (1)$$

The equation of the straight line PQ is

$$y - y_1 = \frac{y_2 - y_1}{x_2 - x_1}(x - x_1). \quad \ldots (2)$$

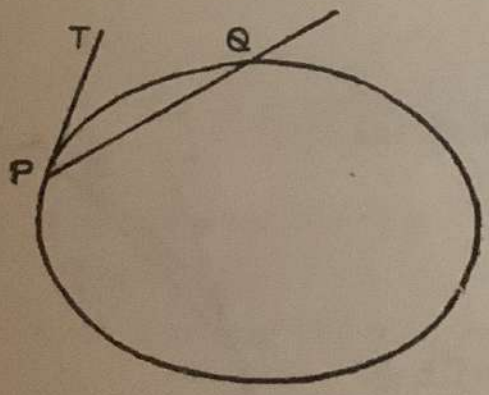


Fig. 21

The points $P(x_1, y_1)$ and $Q(x_2, y_2)$ lie on the ellipse.

Hence $\frac{x_1^2}{a^2} + \frac{y_1^2}{b^2} = 1$ $\ldots (3)$

and $\frac{x_2^2}{a^2} + \frac{y_2^2}{b^2} = 1.$

Subtracting,

$$\frac{x_2^2 - x_1^2}{a^2} + \frac{y_2^2 - y_1^2}{b^2} = 0$$

or, $$\frac{y_2 - y_1}{x_2 - x_1} = -\frac{b^2(x_2 + x_1)}{a^2(y_2 + y_1)}.$$

Substituting this in (2), the equation of the chord of the ellipse becomes

$$y - y_1 = -\frac{b^2(x_2 + x_1)}{a^2(y_2 + y_1)}(x - x_1). \quad \ldots (4)$$

As Q approaches and coincides with P, the chord PQ becomes the tangent at P. Substituting x_1 for x_2 and y_1 for y_2 in (4), the equation of the tangent at P becomes

$$y - y_1 = -\frac{2b^2x_1}{2a^2y_1}(x - x_1)$$

i.e., $\quad \frac{xx_1}{a^2} + \frac{yy_1}{b^2} = \frac{x_1^2}{a^2} + \frac{y_1^2}{b^2} = 1,$ by (3).

$\therefore$ The equation of the tangent at $P(x_1, y_1)$ is

$$\frac{xx_1}{a^2} + \frac{yy_1}{b^2} = 1.$$

Cor. The tangents at the extremities of the major axis, that is, at $(a, 0)$ and $(-a, 0)$ are $x = a$ and $x = -a$ respectively.

The equation of the tangent to the ellipse $\frac{x^2}{a^2} + \frac{y^2}{b^2} = 1$

at the point $(a \cos \theta, b \sin \theta)$ is $\frac{x \cos \theta}{a} + \frac{y \sin \theta}{b} = 1.$

Note. The equation of the tangent may also be obtained by the second method of Art. 5·2.

5·5. Equation of the tangent to the hyperbola $\frac{x^2}{a^2} - \frac{y^2}{b^2} = 1$ at the point (x_1, y_1).

Proceeding exactly in the same way as in the case of ellipse, we get the equation of the tangent to the hyperbola at (x_1, y_1) as

$$\frac{xx_1}{a^2} - \frac{yy_1}{b^2} = 1.$$

The equation of the tangent to the hyperbola $\frac{x^2}{a^2} - \frac{y^2}{b^2} = 1$

at the point $(a \sec \phi, b \tan \phi)$ is

$$\frac{x \sec \phi}{a} - \frac{y \tan \phi}{b} = 1.$$

Cor. The equation of the tangent to the rectangular hyperbola $x^2 - y^2 = a^2$ at the point (x_1, y_1) is $xx_1 - yy_1 = a^2$.

Note. The equation of the tangent to any conic at the point (x_1, y_1) is obtained from the equation of the conic by putting xx_1 for x^2, yy_1 for y^2, $(xy_1 + x_1y)$ for $2xy$, $(x + x_1)$ for $2x$, $(y + y_1)$ for $2y$ and retaining the constant term as it is.

5·6. Condition of tangency of a straight line to a circle.

Let the equation of the circle be $x^2+y^2=a^2$ ... (1)

and that of the straight line be $y=mx+c$. ... (2)

The co-ordinates of the points of intersection of the circle and the line are obtained by solving the two equations. Eliminating y between the two equations, we get

$$x^2+(mx+c)^2=a^2$$

or, $(1+m^2)x^2+2mcx+(c^2-a^2)=0.$

This quadratic in x gives two roots which are the abscissae of the two points of intersection. If the line be a tangent to the circle, it cuts the circle at two coincident points and the above equation should give two equal roots. The condition for that is

$$(2mc)^2-4(1+m^2)(c^2-a^2)=0$$

or, $c^2=a^2(1+m^2)$

or, $c=\pm a\sqrt{1+m^2}$.

This is the required condition of tangency.

Thus there are two tangents to a circle parallel to a given direction.

Second method.

Let the straight line $y=mx+c$ ... (1)

touch the circle $x^2+y^2=a^2$ ... (2)

at the point (x_1, y_1).

The equation of the tangent at (x_1, y_1) to the circle (2) is

$$xx_1+yy_1=a^2. \quad \text{... (3)}$$

Therefore the equations (1) and (3) are identical.

Hence $\frac{x_1}{m}=\frac{y_1}{-1}=-\frac{a^2}{c}$, or, $x_1=-\frac{a^2m}{c}$ and $y_1=\frac{a^2}{c}$.

Since (x_1, y_1) lies on the circle (2), therefore $x_1^2+y_1^2=a^2$

or, $\frac{a^4m^2}{c^2}+\frac{a^4}{c^2}=a^2$ or, $c=\pm a\sqrt{(1+m^2)}$.

This is the required condition of tangency and the point of contact is $\left(-\frac{a^2m}{c}, \frac{a^2}{c}\right)$.

Note. This condition may also be obtained by using the property that the radius of the circle is equal to the distance of the centre of the circle from the tangent line.

5·7. Condition of tangency of a straight line to a parabola.

The points of intersection of the straight line

$$y = mx + c \quad \ldots \quad (1)$$

with the parabola $y^2 = 4ax \quad \ldots \quad (2)$

are obtained by solving these two equations.

Eliminating y from (1) and (2), we get

$$(mx + c)^2 = 4ax$$

or, $$m^2x^2 + (2mc - 4a)x + c^2 = 0. \quad \ldots \quad (3)$$

The roots of (3) are the abscissae of the two points of intersection.

If the line (1) touches the parabola (2), the two points of intersection will be coincident, that is, the roots of (3) are equal.

The condition for equal roots is

$$(2mc - 4a)^2 = 4m^2c^2$$

or, $$16a^2 - 16amc = 0$$

or, $$c = a/m.$$

Thus, for any value of $m \neq 0$, the straight line

$$y = mx + a/m$$

touches the parabola $y^2 = 4ax$.

The point of contact can easily be seen to be at $\left(\frac{a}{m^2}, \frac{2a}{m}\right)$.

Note. This condition may also be obtained by the second method of Art. 5·6.

5·8. Condition of tangency of a straight line to an ellipse.

The points of intersection of the straight line

$$y = mx + c \quad \ldots \quad (1)$$

with the ellipse $\frac{x^2}{a^2} + \frac{y^2}{b^2} = 1 \quad \ldots \quad (2)$

are obtained by solving these two equations.

Eliminating y between these two equations, we get

$$\frac{x^2}{a^2} + \frac{(mx + c)^2}{b^2} = 1$$

or, $$b^2x^2 + a^2(mx + c)^2 = a^2b^2$$

or, $$(a^2m^2 + b^2)x^2 + 2a^2mcx + a^2(c^2 - b^2) = 0.$$

The roots of this equation are the abscissae of the two points of intersection. If the line be a tangent, the two points are coincident and hence the roots of this quadratic will be equal. The condition for equal roots is

$$4m^2c^2a^4 - 4(a^2m^2 + b^2)(a^2c^2 - a^2b^2) = 0$$

or, $$a^2m^2c^2 - (a^2m^2c^2 - a^2m^2b^2 + b^2c^2 - b^4) = 0$$

or, $$c^2 = a^2m^2 + b^2.$$

$$\therefore \quad c = \pm \sqrt{(a^2m^2 + b^2)}.$$

Thus the lines $y = mx \pm \sqrt{(a^2m^2 + b^2)}$ are tangents to the ellipse for any value of m. Hence there are two tangents to an ellipse parallel to any given direction (*i.e.*, for same value of m).

The point of contact of this tangent can easily be verified to be $\left(\frac{-a^2m}{\sqrt{a^2m^2 + b^2}}, \frac{b^2}{\sqrt{a^2m^2 + b^2}}\right)$.

Note. This result may also be obtained by the second method of Art. 5·6.

5·9. Condition of tangency of a straight line to a hyperbola.

Proceeding exactly in the same way as in the case of ellipse, we see that the condition of tangency of the straight line $y = mx + c$ to the hyperbola $\frac{x^2}{a^2} - \frac{y^2}{b^2} = 1$ is

$$c = \pm \sqrt{(a^2m^2 - b^2)}.$$

5·10. Two theorems.

(a) *Two tangents can be drawn from a point to a parabola. If these two tangents be perpendicular to one another, the locus of their point of intersection is the directrix.*

The straight line $y = mx + a/m$ always touches the parabola $y^2 = 4ax$. If this line passes through a given point (α, β), we have

$$\beta = m\alpha + a/m$$

or, $$m^2\alpha - m\beta + a = 0,$$

which being a quadratic in m, has two roots m_1 and m_2 (say). Hence two tangents can be drawn from a point (α, β) to the parabola.

The equations of the tangents are thus

$y = m_1x + a/m_1$ and $y = m_2x + a/m_2$, where $m_1m_2 = a/a$.

If these two tangents be perpendicular to one another, we have

$$m_1m_2 = -1$$

or, $\quad a/a = -1$ whence $a + a = 0$.

Hence the locus of (α, β) is the straight line $x + a = 0$, which is the directrix of the given parabola.

(*b*) *Two tangents can be drawn from a point to a central conic. If these two tangents be perpendicular to one another, the locus of their point of intersection is a circle.* director circle.

The straight line $y = mx + \sqrt{\frac{m^2}{a} + \frac{1}{b}}$ always touches the central conic $ax^2 + by^2 = 1$. If this line passes through a given point (α, β), we have

$$\beta = m\alpha + \sqrt{\frac{m^2}{a} + \frac{1}{b}}$$

or, $$m^2\left(\frac{1}{a} - \alpha^2\right) + 2m\alpha\beta + \left(\frac{1}{b} - \beta^2\right) = 0,$$

which being a quadratic in m, has two roots m_1 and m_2 (say). Hence two tangents can be drawn from (α, β) to the central conic.

The equations of the tangents are

$$y = m_1x + \sqrt{\frac{m_1^2}{a} + \frac{1}{b}} \text{ and } y = m_2x + \sqrt{\frac{m_2^2}{a} + \frac{1}{b}}$$

where $m_1m_2 = \left(\frac{1}{b} - \beta^2\right) \Big/ \left(\frac{1}{a} - \alpha^2\right)$.

If these two tangents be perpendicular of one another, we have

$$m_1m_2 = -1, \text{ giving } \frac{1}{b} - \beta^2 = -\left(\frac{1}{a} - \alpha^2\right).$$

$$\therefore \quad \alpha^2 + \beta^2 = \frac{1}{a} + \frac{1}{b}.$$

Hence the locus of (α, β) is $x^2 + y^2 = \frac{1}{a} + \frac{1}{b}$,

which is a circle.

This is called the *director circle* of the central conic.

5·12. Equation of the pair of tangents from an external point (x_1, y_1) to the circle $x^2+y^2=a^2$.

Let (h, k) be any point on either of the tangents from the point (x_1, y_1) to the circle $x^2+y^2=a^2$.

Now the length of the perpendicular from the centre to the tangent is equal to the radius of the circle. Here the length of the perpendicular from the origin to the line joining (h, k) and (x_1, y_1) must be equal to a, the radius of the circle. The equation of the straight line passing through (h, k) and (x_1, y_1) is

$$y-y_1=\frac{k-y_1}{h-x_1}(x-x_1)$$

or, $$y(h-x_1)-x(k-y_1)+kx_1-hy_1=0. \quad \cdots \quad (1)$$

Hence the length of perpendicular from the origin to (1)

is $$\frac{kx_1-hy_1}{\sqrt{(h-x_1)^2+(k-y_1)^2}}=a$$

so that $(kx_1-hy_1)^2=a^2[(h-x_1)^2+(k-y_1)^2]$.

Hence the locus of (h, k) is

$$(x_1y-xy_1)^2=a^2[(x-x_1)^2+(y-y_1)^2]$$

which is the required pair of tangents.

This may be simplified as

$$x^2(y_1^2-a^2)+y^2(x_1^2-a^2)-a^2(x_1^2+y_1^2)$$
$$=2x_1y_1xy-2a^2x_1x-2a^2y_1y$$

or, $$(x^2+y^2-a^2)(x_1^2+y_1^2-a^2)$$
$$=x^2x_1^2+y^2y_1^2+a^4+2x_1y_1xy-2a^2x_1x-2a^2y_1y$$
$$=(xx_1+yy_1-a^2)^2.$$

Let $S\equiv x^2+y^2-a^2$, $S_1\equiv x_1^2+y_1^2-a^2$

and $T\equiv xx_1+yy_1-a^2$.

Then the equation of the pair of tangents from (x_1, y_1) to the circle is

$$SS_1=T^2.$$

Second method.

For the points of intersection of the circle $x^2+y^2=a^2$ and the straight line $\frac{x-x_1}{l}=\frac{y-y_1}{m}=r$, $l^2+m^2=1$ through the point (x_1, y_1), we have

$$(x_1+lr)^2+(y_1+mr)^2=a^2$$

or, $$r^2+2r(lx_1+my_1)+(x_1^2+y_1^2-a^2)=0.$$

If the straight line be a tangent from (x_1, y_1) to the circle, the two points of intersection will coincide and hence the roots of this quadratic equation in r must be equal.

Therefore $(lx_1 + my_1)^2 = x_1^2 + y_1^2 - a^2$.

Eliminating l, m from the equation of the straight line and the last equation, we have

$$\{x_1(x-x_1) + y_1(y-y_1)\}^2 = (x_1^2 + y_1^2 - a^2)\{(x-x_1)^2 + (y-y_1)^2\}$$

or, $$(xx_1 + yy_1 - a^2)^2 = (x^2 + y^2 - a^2)(x_1^2 + y_1^2 - a^2).$$

This, being satisfied by any point on the tangent drawn through (x_1, y_1), gives the equation of the pair of tangents from (x_1, y_1) to the circle.

Cor. Proceeding in the same way, we obtain the equation of the pair of tangents from (x_1, y_1) to the circle

$$x^2 + y^2 + 2gx + 2fy + c = 0 \text{ as } SS_1 = T^2,$$

where $$S \equiv x^2 + y^2 + 2gx + 2fy + c,$$

$$S_1 \equiv x_1^2 + y_1^2 + 2gx_1 + 2fy_1 + c$$

and $$T \equiv xx_1 + yy_1 + g(x + x_1) + f(y + y_1) + c.$$

5·13. Equation of the pair of tangents from (x_1, y_1) to the parabola $y^2 = 4ax$.

For the points of intersection of the parabola $y^2 = 4ax$ and the straight line

$$\frac{x - x_1}{l} = \frac{y - y_1}{m} = r, \quad l^2 + m^2 = 1$$

through the point (x_1, y_1), we have

$$(y_1 + mr)^2 = 4a(x_1 + lr)$$

or, $$m^2r^2 - 2r(2al - my_1) + (y_1^2 - 4ax_1) = 0.$$

If the straight line be a tangent from (x_1, y_1) to the parabola, the two points of intersection will coincide and hence the roots of this quadratic equation in r must be equal.

Therefore $$(2al - my_1)^2 = m^2(y_1^2 - 4ax_1)$$

or, $$al^2 - lmy_1 + m^2x_1 = 0.$$

Eliminating l, m from the equation of the straight line and the last equation, we have

$$a(x - x_1)^2 - (x - x_1)(y - y_1)y_1 + (y - y_1)^2 x_1 = 0$$

or, $$(y^2 - 4ax)(y_1^2 - 4ax_1) = \{yy_1 - 2a(x + x_1)\}^2.$$

This, being satisfied by any point on the tangent drawn through (x_1, y_1), gives the equation of the pair of tangents from (x_1, y_1) to the parabola.

Thus the equation of the pair of tangents from (x_1, y_1) to the parabola is $SS_1 = T^2$,

where $S \equiv y^2 - 4ax$, $S_1 \equiv y_1^2 - 4ax_1$

and $T \equiv yy_1 - 2a(x + x_1)$.

5·14. Equation of the pair of tangents from (x_1, y_1) to the ellipse $\frac{x^2}{a^2} + \frac{y^2}{b^2} = 1$.

The equation of a straight line through the point (x_1, y_1) is $\frac{x - x_1}{l} = \frac{y - y_1}{m} = r$, where $l^2 + m^2 = 1$.

For the intersection of this straight line and the ellipse, we have

$$\frac{(x_1 + lr)^2}{a^2} + \frac{(y_1 + mr)^2}{b^2} = 1$$

or, $$r^2\left(\frac{l^2}{a^2} + \frac{m^2}{b^2}\right) + 2r\left(\frac{lx_1}{a^2} + \frac{my_1}{b^2}\right) + \left(\frac{x_1^2}{a^2} + \frac{y_1^2}{b^2} - 1\right) = 0.$$

If the straight line touches the ellipse, the roots of this quadratic equation in r will be equal.

Hence $$\left(\frac{lx_1}{a^2} + \frac{my_1}{b^2}\right)^2 = \left(\frac{l^2}{a^2} + \frac{m^2}{b^2}\right)\left(\frac{x_1^2}{a^2} + \frac{y_1^2}{b^2} - 1\right).$$

Eliminating l, m from the equation of the straight line and the above equation, we have

$$\left\{\frac{x_1(x - x_1)}{a^2} + \frac{y_1(y - y_1)}{b^2}\right\}^2 = \left\{\frac{(x - x_1)^2}{a^2} + \frac{(y - y_1)^2}{b^2}\right\}\left(\frac{x_1^2}{a^2} + \frac{y_1^2}{b^2} - 1\right)$$

which may be written as $SS_1 = T^2$,

where $$S \equiv \frac{x^2}{a^2} + \frac{y^2}{b^2} - 1, \quad S_1 \equiv \frac{x_1^2}{a^2} + \frac{y_1^2}{b^2} - 1$$

and $$T \equiv \frac{xx_1}{a^2} + \frac{yy_1}{b^2} - 1.$$

This, being satisfied by any point on the tangent drawn through (x_1, y_1), gives the equation of the pair of tangents from (x_1, y_1) to the ellipse.

This, being satisfied by any point on the tangent drawn through (x_1, y_1), gives the equation of the pair of tangents from (x_1, y_1) to the parabola.

Thus the equation of the pair of tangents from (x_1, y_1) to the parabola is $SS_1 = T^2$,

where $S \equiv y^2 - 4ax$, $S_1 \equiv y_1^2 - 4ax_1$

and $T \equiv yy_1 - 2a(x + x_1)$.

5·14. Equation of the pair of tangents from (x_1, y_1) to the ellipse $\frac{x^2}{a^2} + \frac{y^2}{b^2} = 1$.

The equation of a straight line through the point (x_1, y_1) is $\frac{x - x_1}{l} = \frac{y - y_1}{m} = r$, where $l^2 + m^2 = 1$.

For the intersection of this straight line and the ellipse, we have

$$\frac{(x_1 + lr)^2}{a^2} + \frac{(y_1 + mr)^2}{b^2} = 1$$

or, $$r^2\left(\frac{l^2}{a^2} + \frac{m^2}{b^2}\right) + 2r\left(\frac{lx_1}{a^2} + \frac{my_1}{b^2}\right) + \left(\frac{x_1^2}{a^2} + \frac{y_1^2}{b^2} - 1\right) = 0.$$

If the straight line touches the ellipse, the roots of this quadratic equation in r will be equal.

Hence $$\left(\frac{lx_1}{a^2} + \frac{my_1}{b^2}\right)^2 = \left(\frac{l^2}{a^2} + \frac{m^2}{b^2}\right)\left(\frac{x_1^2}{a^2} + \frac{y_1^2}{b^2} - 1\right).$$

Eliminating l, m from the equation of the straight line and the above equation, we have

$$\left\{\frac{x_1(x - x_1)}{a^2} + \frac{y_1(y - y_1)}{b^2}\right\}^2 - \left\{\frac{(x - x_1)^2}{a^2} + \frac{(y - y_1)^2}{b^2}\right\}\left(\frac{x_1^2}{a^2} + \frac{y_1^2}{b^2} - 1\right)$$

which may be written as $SS_1 = T^2$,

where $$S \equiv \frac{x^2}{a^2} + \frac{y^2}{b^2} - 1, \quad S_1 \equiv \frac{x_1^2}{a^2} + \frac{y_1^2}{b^2} - 1$$

and $$T \equiv \frac{xx_1}{a^2} + \frac{yy_1}{b^2} - 1.$$

This, being satisfied by any point on the tangent drawn through (x_1, y_1), gives the equation of the pair of tangents from (x_1, y_1) to the ellipse.

Cor. Proceeding in the same way, we get the equation of the pair of tangents from (x_1, y_1) to the hyperbola

$$\frac{x^2}{a^2} - \frac{y^2}{b^2} = 1 \quad \text{as} \quad SS_1 = T^2,$$

where $$S \equiv \frac{x^2}{a^2} - \frac{y^2}{b^2} - 1, \quad S_1 \equiv \frac{x_1^2}{a^2} - \frac{y_1^2}{b^2} - 1$$

and $$T \equiv \frac{xx_1}{a^2} - \frac{yy_1}{b^2} - 1.$$

5·15. Chord of contact of tangents.

The *chord of contact* of tangents from an external point is the straight line joining the points of contact of the pair of tangents from the external point to the conic.

5·16. Equation of the chord of contact of tangents from (x_1, y_1) to the circle $x^2 + y^2 = a^2$.

Let $A(x_2, y_2)$ and $B(x_3, y_3)$ be the points of contact of the tangents from $P(x_1, y_1)$ to the circle $x^2 + y^2 = a^2$.

The straight line AB is the chord of contact.

Now the equation of the tangent PA at $A(x_2, y_2)$ is

$$xx_2 + yy_2 = a^2.$$

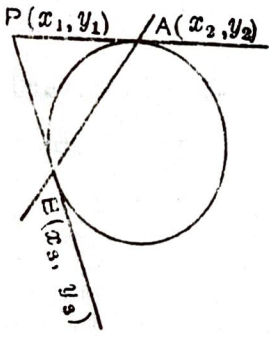


Fig. 23

Since it passes through $P(x_1, y_1)$, we have

$$x_1x_2 + y_1y_2 = a^2. \quad \ldots (1)$$

Similarly, since the tangent PB at $B(x_3, y_3)$ passes through $P(x_1, y_1)$, we have

$$x_1x_3 + y_1y_3 = a^2. \quad \ldots (2)$$

The relations (1) and (2) show that both (x_2, y_2) and (x_3, y_3) satisfy the equation

$$xx_1 + yy_1 = a^2$$

which represents a straight line.

This is therefore the equation of the chord of contact.

Cor. The equation of the chord of contact of tangents from (x_1, y_1) to the circle $x^2 + y^2 + 2gx + 2fy + c = 0$ is

$$xx_1 + yy_1 + g(x + x_1) + f(y + y_1) + c = 0.$$

5·17. Equation of the chord of contact of tangents from (x_1, y_1) to the parabola $y^2 = 4ax$.

Let $A(x_2, y_2)$ and $B(x_3, y_3)$ be the points of contact of the tangents from $P(x_1, y_1)$ to the parabola $y^2 = 4ax$.

The straight line AB is the chord of contact.

Now the equation of the tangent PA at $A(x_2, y_2)$ is

$$yy_2 = 2a(x + x_2).$$

Since it passes through $P(x_1, y_1)$, we have

$$y_1y_2 = 2a(x_1 + x_2). \quad \cdots (1)$$

Similarly, since the tangent PB at $B(x_3, y_3)$ passes through $P(x_1, y_1)$, we have

$$y_1y_3 = 2a(x_1 + x_3). \quad \cdots (2)$$

The relations (1) and (2) show that both (x_2, y_2) and (x_3, y_3) lie on the straight line $yy_1 = 2a(x + x_1)$.

This is the equation of the chord of contact AB.

5·18. Equation of the chord of contact of tangents from (x_1, y_1) to the ellipse $\frac{x^2}{a^2} + \frac{y^2}{b^2} = 1$.

Let $A(x_2, y_2)$ and $B(x_3, y_3)$ be the points of contact of the tangents from $P(x_1, y_1)$ to the ellipse $\frac{x^2}{a^2} + \frac{y^2}{b^2} = 1$.

The equation of the tangent PA at $A(x_2, y_2)$ is

$$\frac{xx_2}{a^2} + \frac{yy_2}{b^2} = 1.$$

Since it passes through $P(x_1, y_1)$, we have

$$\frac{x_1x_2}{a^2} + \frac{y_1y_2}{b^2} = 1. \quad \cdots (1)$$

Similarly, since the tangent at B passes though P, we have

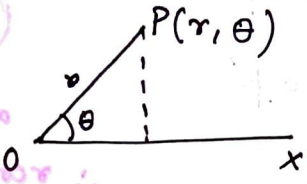
$$\frac{x_1x_3}{a^2} + \frac{y_1y_3}{b^2} = 1. \quad \cdots (2)$$

The relations (1) and (2) show that the two points A and B lie on the straight line $\frac{xx_1}{a^2} + \frac{yy_1}{b^2} = 1$.

This is the equation of the chord of contact AB.

POLAR EQUATION

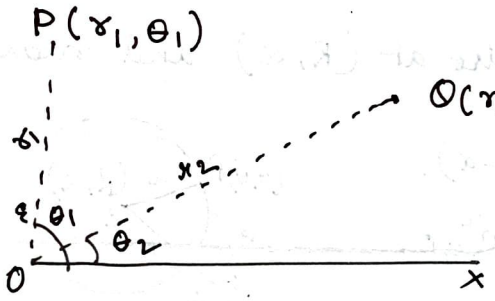
origin or pole : O & initial line : OX



$x = r\cos\theta$ $\quad y = r\sin\theta$. $\quad \therefore r^2 = x^2 + y^2$

and $\tan\theta = y/x$.

1. Distance between 2 points :

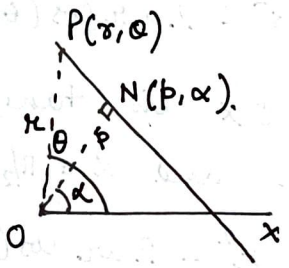


From trigonometry $PQ^2 = PO^2 + OQ^2 - 2\,PO\cdot OQ\cos(\theta_1 - \theta_2)$

or, $PQ^2 = r_1^2 + r_2^2 - 2r_1r_2\cos(\theta_1 - \theta_2)$

2. Equation of a straight line :

$\cos(\theta - \alpha) = \frac{p}{r} \Rightarrow r\cos(\theta - \alpha) = p.$



1. If the st. line is ⊥ to OX then ON co-incides with OX and hence $\alpha = 0$.

$\therefore r\cos\theta = p.$

2. If the st. ~~line~~ line is ∥ to OX then $\alpha = \pi/2$

$\therefore r\cos(\theta - \pi/2) = p$ or, $r\cos(\pi/2 - \theta) = p \Rightarrow r\sin\theta = p.$

3. If ~~p=0~~ the st. line passes through the pole, then $p = 0$

$\therefore r\cos(\theta - \alpha) = 0 \Rightarrow \theta - \alpha = \pi/2 \Rightarrow \theta =$ constant.

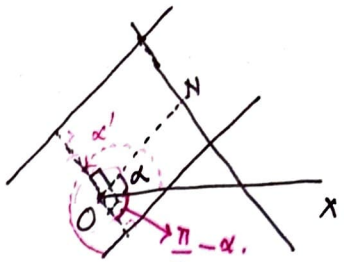
4. ∥ st. lines : $r\cos(\theta-\alpha)=p$ & $r\cos(\theta-\alpha)=p'$

5. ⊥ st. lines : $r\cos(\theta-\alpha)=p$ &

$$r\cos\left(\theta-\alpha-\frac{\pi}{2}\right)=p'$$

or, $r\cos(\theta-\alpha')=p'$

where $\alpha'-\alpha=\pi/2$



$\alpha' = 2\pi-(\frac{\pi}{2}-\alpha)$

$\therefore r\cos\left(\theta-2\pi+\frac{\pi}{2}-\alpha\right)=p'$

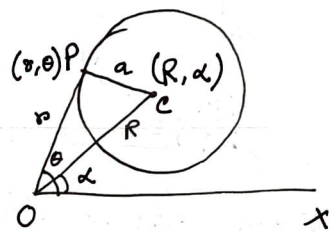
$\Rightarrow r\cos\left(\theta+\frac{\pi}{2}-\alpha-2\pi\right)=p'$

$\Rightarrow r\cos(\theta+\pi/2-\alpha)=p'$

6. $r\cos(\theta-\alpha)=p \Rightarrow \frac{1}{r}=\frac{\cos\alpha}{p}\cos\theta+\frac{\sin\alpha}{p}\sin\theta.$

$\Rightarrow \frac{1}{r}=A\cos\theta+B\sin\theta.$

Equation of a circle : centre at (R,α) and radius $=a$.



here $a^2=r^2+R^2-2rR\cos(\theta-\alpha)$.

$\Rightarrow r^2-2rR\cos(\theta-\alpha)+R^2-a^2=0.$

1. If OX passes through C then $\alpha=0$.

$\therefore r^2-2rR\cos\theta+R^2-a^2=0.$

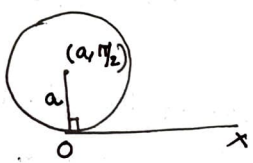
2. If O lies on the circle then $R=a$.

$\therefore r^2-2ar\cos(\theta-\alpha)=0. \Rightarrow r=2a\cos(\theta-\alpha).$

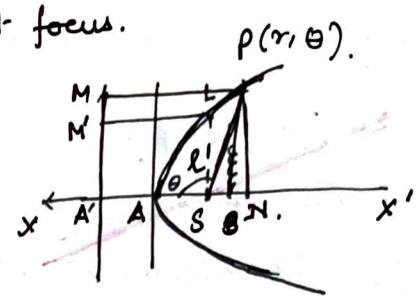
3. If OX is tangent to the circle at O then $R=a$ and $\alpha=\pi/2$

$\therefore r^2-2ar\cos(\theta-\pi/2)=0$

$\Rightarrow r=2a\sin\theta.$



Equation of a conic with pole at focus.



Let eccentricity be e and length of semi latus rectum $= l$.

$\frac{PS}{PM} = e$ and $\frac{LS}{LM'} = e$.

Let $P(r, \theta)$. Then, $PS = e\,PM \Rightarrow r = e[N'A']$.

$\Rightarrow r = e(N'S + SA') = e[r\cos(\pi - \theta) + SA']$.

Now, $SA' = LM'$ & $LS = l$ $\therefore SA' = l/e$.

$\therefore r = -er\cos\theta + l$

$\Rightarrow 1 = -e\cos\theta + \frac{l}{r} \Rightarrow \frac{l}{r} = 1 + e\cos\theta$.

1. If $e = 1$ then parabola.
   If $e < 1$ then ellipse.
   If $e > 1$ then hyperbola.

2. If the initial line is taken along Sx' then $\angle PSx' = \theta$.

$\therefore r = e\left[r\cos\theta + \frac{l}{e}\right] \Rightarrow \frac{l}{r} = 1 - e\cos\theta$.

3. If the axis of the conic is inclined at angle α to the initial line. Then $\angle xSP = \theta - \alpha$. [If the coloured line in the above dig. is the initial line].

$[\because \angle PSN = \pi - (\theta - \alpha)]$

& the eqn of the conic becomes

$\frac{l}{r} = 1 + e\cos(\theta - \alpha)$.

Equation of chord to a conic.

Let $P(r_1, \alpha-\beta)$ & $Q(r_2, \alpha+\beta)$ be the end points of the chord to the conic

$$\frac{l}{r} = 1 + e\cos\theta. \quad \text{—(1)}$$

Let us assume the eq^n of PQ be $\frac{l}{r} = A\cos\theta + B\sin\theta$. —(2)

P & Q lie on (1) & (2).

$$\therefore \frac{l}{r_1} = 1 + e\cos(\alpha-\beta) \;;\; \frac{l}{r_1} = A\cos(\alpha-\beta) + B\sin(\alpha-\beta).$$

$$\frac{l}{r_2} = 1 + e\cos(\alpha+\beta) \;;\; \frac{l}{r_2} = A\cos(\alpha+\beta) + B\sin(\alpha+\beta).$$

Solving for A & B we have

$$(A - e)\cos(\alpha-\beta) + B\sin(\alpha-\beta) - 1 = 0$$
$$(A - e)\cos(\alpha+\beta) + B\sin(\alpha+\beta) - 1 = 0.$$

$$\therefore \frac{A-e}{-\cos(\alpha-\beta)+\cos(\alpha+\beta)} = \frac{B}{-\cos(\alpha+\beta)+\cos(\alpha-\beta)} = \frac{1}{\cos(\alpha-\beta)\sin(\alpha+\beta) - \sin(\alpha-\beta)\cos(\alpha+\beta)}$$

$$\Rightarrow A - e = \frac{2\cos\alpha\sin\beta}{\sin 2\beta} = \cos\alpha\sec\beta.$$

$$B = \frac{2\sin\alpha\sin\beta}{\sin 2\beta} = \sin\alpha\sec\beta.$$

$$\therefore \text{from (2)} \quad \frac{l}{r} = (e + \cos\alpha\sec\beta)\cos\theta + \sin\alpha\sec\beta\sin\theta$$
$$= e\cos\theta + \sec\beta\left[\cos\alpha\cos\theta + \sin\alpha\sin\theta\right]$$
$$= e\cos\theta + \sec\beta\cos(\theta-\alpha). \quad \text{—(3).}$$

Eq^n of tangent to a conic at (α).

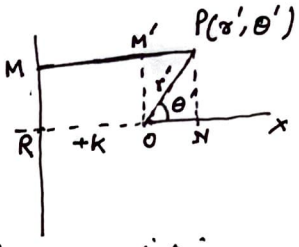
If P & Q are the same then, $\beta = 0$.

$\therefore$ from (3) $\frac{l}{r} = e\cos\theta + \cos(\theta-\alpha)$ is the eq^n of the tangent.

Show that the locus of a point whose distance from the pole is equal to its distance from the straight line $r\cos\theta + k = 0$ is $2r\sin^2\frac{\theta}{2} = k$.

4(e). straight line : $r\cos\theta + k = 0$

here $\alpha = 0$. i.e st. line is $\perp$ to OX. and on the l.h.s. of OX in the opp. direction.



Let $P(r', \theta')$ be any point. By the given conditions

$OP = PM$. —(1)

$PM = PM' + M'M = ON + OR. = r'\cos\theta' + k$

$\therefore$ from (1) ~~$k + r'\cos\theta'$~~ $r' = k + r'\cos\theta'$

$\Rightarrow (1 - \cos\theta')r' = k \Rightarrow \left(1 - \cos^2\frac{\theta'}{2} + \sin^2\frac{\theta'}{2}\right)r' = k$

$\Rightarrow 2r'\sin^2\frac{\theta'}{2} = k.$

Thus the locus of (r', θ') is $2r\sin^2\frac{1}{2}\theta = k$.

5. Find the condition that the three straight lines

$r\cos(\theta - \alpha) = a$; $r\cos(\theta - \beta) = b$; $r\cos(\theta - \gamma) = c$

—(1) —(2) —(3)

may meet at a point

Sol^n. The 3 st. lines ~~are~~ meet at a point if solving r, θ from (1) & (2) satisfies (3).

$r\cos\theta\cos\alpha + r\sin\theta\sin\alpha = a$

$\Rightarrow (r\cos\alpha)\cos\theta + (r\sin\alpha)\sin\theta - a = 0$

Simi. $(r\cos\beta)\cos\theta + (r\sin\beta)\sin\theta - b = 0$

Solving for $\cos\theta$ & $\sin\theta$ we have

$$\frac{\cos\theta}{-rb\sin\alpha + ar\sin\beta} = \frac{\sin\theta}{-ar\cos\beta + br\cos\alpha} = \frac{1}{r^2\cos\alpha\sin\beta - r^2\sin\alpha\cos\beta}$$

$$\Rightarrow \cos\theta = \frac{-b\sin\alpha + a\sin\beta}{r\sin(\beta - \alpha)} \; ; \; \sin\theta = \frac{b\cos\alpha - a\cos\beta}{r\sin(\beta - \alpha)}.$$

Subst^n in (3)

$r\cos\theta\cos\gamma + r\sin\theta\sin\gamma = c$

$\Rightarrow (-b\sin\alpha + a\sin\beta)\cos\gamma + (b\cos\alpha - a\cos\beta)\sin\gamma = c\sin(\beta - \alpha)$

$\Rightarrow -b\sin(\alpha - \gamma) + a\sin(\beta - \gamma) - c\sin(\beta - \alpha) = 0$

Problem: Find the eqⁿ of the tangent at $\theta = \alpha$ to the circle which passes through the pole.

Solⁿ. Let the equation of the circle be

$$r = 2a\cos(\theta - \gamma) \quad \text{where center is at } (a, \gamma) \text{ and radius} = a. \quad \text{——— (1)}$$

Let $P(\alpha - \beta)$ and $Q(\alpha + \beta)$ be two pts. on (1)

Let eqⁿ of PQ be

$$\frac{2a}{r} = A\cos\theta + B\sin\theta \quad \text{——— (2)}.$$

P & Q both lie on (1) & (2).

$$\therefore r_1 = 2a\cos(\alpha - \beta - \gamma) \quad ; \quad \frac{2a}{r_1} = A\cos(\alpha - \beta) + B\sin(\alpha - \beta).$$

and $r_2 = 2a\cos(\alpha + \beta - \gamma) \quad ; \quad \frac{2a}{r_2} = A\cos(\alpha + \beta) + B\sin(\alpha + \beta).$

Solving for A & B.

$$\frac{1}{\cos(\alpha - \beta - \gamma)} = A\cos(\alpha - \beta) + B\sin(\alpha - \beta)$$

$$\frac{1}{\cos(\alpha + \beta - \gamma)} = A\cos(\alpha + \beta) + B\sin(\alpha + \beta).$$

OR. $$\frac{\cos(\alpha + \beta)}{\cos(\alpha - \beta - \gamma)} - \frac{\cos(\alpha - \beta)}{\cos(\alpha + \beta - \gamma)} = B\left[\sin(\alpha - \beta)\cos(\alpha + \beta) - \sin(\alpha + \beta)\cos(\alpha - \beta)\right]$$

$$\Rightarrow B.\sin 2\beta = \frac{\cos(\alpha - \beta)\cos(\alpha - \beta - \gamma) - \cos(\alpha + \beta)\cos(\alpha + \beta - \gamma)}{\cos(\alpha - \beta - \gamma)\cos(\alpha + \beta - \gamma)}.$$

$$\Rightarrow B\sin 2\beta = \frac{\frac{1}{2}\left[\cos[2(\alpha - \beta) - \gamma] + \cos\gamma\right] - \frac{1}{2}\left[\cos(2(\alpha + \beta) - \gamma) + \cos\gamma\right]}{\frac{1}{2}\left[\cos 2(\alpha - \gamma) + \cos 2\beta\right]}$$

$$= \frac{\cos(2\alpha - 2\beta - \gamma) - \cos(2\alpha + 2\beta - \gamma)}{\cos(2\alpha - 2\gamma) + \cos 2\beta}.$$

or, $B \sin 2\beta = \dfrac{2 \sin(2\alpha - \gamma) \sin 2\beta}{K}$ where $K = \cos(2\alpha - 2\gamma) + \cos 2\beta$.

$\Rightarrow B = 2 \sin(2\alpha - \gamma)/K$.

Again

$$A\left[\cos(\alpha-\beta)\sin(\alpha+\beta) - \cos(\alpha+\beta)\sin(\alpha-\beta)\right]$$

$$= \frac{\sin(\alpha+\beta)\cos(\alpha+\beta-\gamma) - \sin(\alpha-\beta)\cos(\alpha-\beta-\gamma)}{\frac{1}{2}K}.$$

$$A \sin 2\beta = \frac{\frac{1}{2}\left[\sin(2\alpha+2\beta-\gamma) + \sin\gamma\right] - \frac{1}{2}\left[\sin(2\alpha-2\beta-\gamma) + \sin\gamma\right]}{\frac{1}{2}K}.$$

$$\Rightarrow A \sin 2\beta = \frac{\sin(2\alpha+2\beta-\gamma) - \sin(2\alpha-2\beta-\gamma)}{K}$$

$$= \frac{2\cos(2\alpha-\gamma)\sin 2\beta}{K}.$$

i.e $A = 2\cos(2\alpha-\gamma)/K$.

Then eq^{n} of tangent is at $\theta = \alpha$ i.e $\beta = 0$ is

$$\frac{2a}{\mu} = \frac{2\cos(2\alpha-\gamma)}{1+\cos 2(\alpha-\gamma)}\cos\theta + \frac{2\sin(2\alpha-\gamma)}{1+\cos 2(\alpha-\gamma)}\sin\theta.$$

$$\Rightarrow \frac{a}{\gamma} = \frac{\cos(\theta - 2\alpha + \gamma)}{1+\cos(2\alpha-2\gamma)} \text{ or } \frac{\cos(2\alpha-\gamma-\theta)}{1+\cos(2\alpha-2\gamma)}.$$

Find the polar eqn of the st. line joining two points on the parabola

16 (a) $\frac{2a}{r} = 1 + \cos\theta$ — (1). with $(\alpha-\beta)$ and $(\alpha+\beta)$ as their vectorial angles.

If $(r_1, \alpha-\beta)$ and $(r_2, \alpha+\beta)$ lies on (1) then

$$\frac{2a}{r_1} = 1+\cos(\alpha-\beta) \quad \& \quad \frac{2a}{r_2} = 1+\cos(\alpha+\beta).$$

Let Eqn of the st. line joining the two pts be

$$\frac{1}{r} = A\cos\theta + B\sin\theta.$$

Then $$\frac{1}{r_1} = A\cos(\alpha-\beta) + B\sin(\alpha-\beta)$$

$$\frac{1}{r_2} = A\cos(\alpha+\beta) + B\sin(\alpha+\beta).$$

$$\frac{\cos(\alpha+\beta)}{r_1} = A\cos(\alpha-\beta)\cos(\alpha+\beta) + B\cos(\alpha+\beta)\sin(\alpha-\beta).$$

$$\frac{\cos(\alpha-\beta)}{r_2} = A\cos(\alpha-\beta)\cos(\alpha+\beta) + B\cos(\alpha-\beta)\sin(\alpha+\beta).$$

$$\Rightarrow \frac{\cos(\alpha+\beta)}{r_1} - \frac{\cos(\alpha-\beta)}{r_2} = B\left[\cos(\alpha+\beta)\sin(\alpha-\beta) - \cos(\alpha-\beta)\sin(\alpha+\beta)\right].$$

$$\Rightarrow \cos(\alpha+\beta)\left[\frac{1+\cos(\alpha-\beta)}{2a}\right] - \cos(\alpha-\beta)\left[\frac{1+\cos(\alpha+\beta)}{2a}\right]$$

$$= B\left[\sin\{\alpha-\beta-\alpha-\beta\}\right].$$

$$\Rightarrow \frac{1}{2a}\left[\cos(\alpha+\beta) - \cos(\alpha-\beta)\right] = -B\sin 2\beta$$

$$\Rightarrow \frac{1}{2a} \times 2\sin\alpha\sin\beta = 2B\sin\beta\cos\beta. \Rightarrow B = \frac{\sin\alpha}{2a\cos\beta}.$$

Again. $$\frac{\sin(\alpha+\beta)}{r_1} = A\cos(\alpha-\beta)\sin(\alpha+\beta) + B\sin(\alpha-\beta)\sin(\alpha+\beta)$$

$$\frac{\sin(\alpha-\beta)}{r_2} = A\cos(\alpha+\beta)\sin(\alpha-\beta) + B\sin(\alpha+\beta)\sin(\alpha-\beta)$$

$$\text{or, } \frac{\sin(\alpha+\beta)}{r_1} - \frac{\sin(\alpha-\beta)}{r_2} = A\left[\cos(\alpha-\beta)\sin(\alpha+\beta) - \cos(\alpha+\beta)\sin(\alpha-\beta)\right].$$

$$\Rightarrow \sin(\alpha+\beta)\left[\frac{1+\cos(\alpha-\beta)}{2a}\right] - \sin(\alpha-\beta)\left[\frac{1+\cos(\alpha+\beta)}{2a}\right]$$

$$= A\sin\left[\alpha+\beta-\alpha+\beta\right].$$

$$\Rightarrow \frac{1}{2a}\left[\sin(\alpha+\beta) - \sin(\alpha-\beta)\right] + \frac{1}{2a}\left[\sin(\alpha+\beta)\cos(\alpha-\beta) - \sin(\alpha-\beta)\cos(\alpha+\beta)\right].$$

$$= A\sin 2\beta.$$

$$\Rightarrow \frac{1}{2a}\left[2\cos\alpha\sin\beta\right] + \frac{1}{2a}\sin 2\beta = A\sin 2\beta.$$

$$\Rightarrow \frac{1}{2a}\left[\frac{2\sin\beta\cos\alpha}{2\sin\beta\cos\beta}\right] + \frac{1}{2a} = A.$$

$$\Rightarrow A = \frac{1}{2a}\left[1 + \frac{\cos\alpha}{\cos\beta}\right].$$

$\therefore$ Eq^n of st. line is

$$\frac{1}{r} = \frac{1}{2a}\left(1+\frac{\cos\alpha}{\cos\beta}\right)\cos\theta + \frac{1}{2a}\frac{\sin\alpha}{\cos\beta}\sin\theta$$

$$\Rightarrow \frac{2a}{r} = \cos\theta + \sec\beta\cos(\theta-\alpha).$$

18 (e.) If the straight line (1) touches the parabola (2), then show that $p = \frac{1}{2}l\sec\alpha$.

$$r\cos(\theta-\alpha) = p \quad \text{—— (1)}.$$

$$\frac{l}{r} = 1+\cos\theta \quad \text{—— (2)}.$$

Eq^n of tangent to (2) at 'β' is $\frac{l}{r} = \cos\theta + \cos(\theta-\beta)$.

$$\text{or, } \frac{l}{r} = (\cos\beta+1)\cos\theta + \sin\beta\sin\theta. \quad \text{—— (3)}.$$

from (1) $\frac{p}{r} = \cos\theta\cos\alpha + \sin\theta\sin\alpha$ —— (4) Since (3) & (4) are identical $\therefore$

$$\frac{l}{p} = \frac{1+\cos\beta}{\cos\alpha} = \frac{\sin\beta}{\sin\alpha}.$$

$$\sin\beta = \frac{l}{p}\sin\alpha \quad \& \quad \cos\beta = \frac{l}{p}\cos\alpha - 1 \quad *$$

* 18(e)

$\sin^2\beta + \cos^2\beta = 1$

$\Rightarrow \frac{l^2}{p^2}\sin^2\alpha + \left(\frac{l}{p}\cos\alpha - 1\right)^2 = 1$

$\Rightarrow \frac{l^2}{p^2} - 2\frac{l}{p}\cos\alpha + 1 = 1 \Rightarrow \frac{l}{p}\left(\frac{l}{p} - 2\cos\alpha\right) = 0$

$\Rightarrow 1/p = 2\cos\alpha \Rightarrow p = \frac{l}{2}\sec\alpha.$

Show that the equation of the circle which passes through the focus of the parabola $\frac{2a}{r} = 1 + \cos\theta$ and touches it at the point $\theta = \alpha$ is given by $r\cos^3\frac{\alpha}{2} = a\cos\left(\theta - \frac{3}{2}\alpha\right)$.

20 (a). Eqn of circle which passes through the pole is

$r = 2a'\cos(\theta - \gamma')$. where centre is at (a', γ'). & radius is a'.
——— (1)

Eqn of parabola : $\frac{2a}{r} = 1 + \cos\theta$. ——— (2)

Eqn of tangent to this parabola at $\theta = \alpha$ is

$$\frac{2a}{r} = \cos\theta + \cos(\theta - \alpha). \quad \text{——— (3).}$$

Eqn of tangent to (1) at $\theta = \alpha$ is

$$\frac{2a'}{r} = \frac{2\cos(2\alpha - \gamma)}{1 + \cos(2\alpha - 2\gamma)}\cos\theta + \frac{2\sin(2\alpha - \gamma)}{1 + \cos(2\alpha - 2\gamma)}\sin\theta.$$
——— (4).

If (1) & (2) have a common tangent at $\theta = \alpha$ then (3) & (4) must be identical.

from (3) $\frac{2a}{r} = \cos\theta(1 + \cos\alpha) + \sin\theta\sin\alpha$.
——— (5).

from (4) & (5) $\frac{a'}{a} = \frac{2\cos(2\alpha - \gamma)}{[1 + \cos\{2(\alpha - \gamma)\}][1 + \cos\alpha]}$

$$= \frac{2\sin(2\alpha - \gamma)}{[1 + \cos(2\alpha - 2\gamma)]\sin\alpha}$$

$\therefore \frac{2\cos(2\alpha - \gamma)}{1 + \cos\alpha} = \frac{2\sin(2\alpha - \gamma)}{\sin\alpha}$.

$\Rightarrow \sin\alpha\cos(2\alpha - \gamma) - \cos\alpha\sin(2\alpha - \gamma) = \sin(2\alpha - \gamma)$

$\Rightarrow \sin[\alpha - 2\alpha + \gamma] = \sin(2\alpha - \gamma)$.

$\Rightarrow -\alpha + \gamma = 2\alpha - \gamma \Rightarrow 2\gamma = 3\alpha \Rightarrow \gamma = \frac{3\alpha}{2}$.

$$\therefore \frac{a'}{a} = \frac{2\cos\left(2\alpha - \frac{3}{2}\alpha\right)}{\left[1 + \cos(2\alpha - 3\alpha)\right](\cos\alpha + 1)}$$

$$\Rightarrow a' = \frac{2a\cos\left(\frac{\alpha}{2}\right)}{(1+\cos\alpha)\cdot(1+\cos\alpha)} = \frac{2a\cos(\alpha/2)}{2\cos^2\frac{\alpha}{2}\cdot 2\cos^2\frac{\alpha}{2}}$$

$$= \frac{a}{2}\cos^{-3}\alpha/2$$

$\therefore$ eqn of circle is $r = \frac{a}{\cos^3\frac{\alpha}{2}}\cdot\cos\left(\theta - \frac{3\alpha}{2}\right)$

$$\Rightarrow r\cos^3\alpha/2 = a\cos\left(\theta - \frac{3}{2}\alpha\right).$$

1. A circle passes through the focus S of a conic and meets it in four points whose distances from S are r_1, r_2, r_3 and r_4. Prove that $\frac{1}{r_1} + \frac{1}{r_2} + \frac{1}{r_3} + \frac{1}{r_4} = \frac{2}{l}$ where $2l$ is the latus-rectum of the conic.

Soln: Let the eqn of the conic be $\frac{l}{r} = 1 - e\cos\theta$ ——(1)

and circle be $r = 2a\cos(\theta - \alpha)$ ——(2) where the centre C is at (a, α); a being the radius.

Solving (1) and (2) we have

$① \rightarrow e\cos\theta = 1 - \frac{l}{r} \Rightarrow \cos\theta = \frac{1}{e}\left(1 - \frac{l}{r}\right)$.

and, $② \rightarrow r = 2a\cos\theta\cos\alpha + 2a\sin\theta\sin\alpha$

$\Rightarrow (r - 2a\cos\alpha\cos\theta)^2 = 4a^2\sin^2\alpha(1 - \cos^2\theta)$.

$\Rightarrow \left\{r - 2a\cos\alpha \cdot \frac{1}{e}\left(1 - \frac{l}{r}\right)\right\}^2 = 4a^2\sin^2\alpha\left\{1 - \frac{1}{e^2}\left(1 - \frac{l}{r}\right)^2\right\}$

$\Rightarrow \left\{er - 2a\cos\alpha(r - l)\right\}^2 = 4a^2\sin^2\alpha\left\{e^2r^2 - (r-l)^2\right\}$

$\Rightarrow e^2r^4 - 4ae r^2\cos\alpha(r-l) + 4a^2\cos^2\alpha(r-l)^2$

$= 4a^2\sin^2\alpha(e^2r^2 - r^2 - l^2 + 2rl)$

$\Rightarrow e^2r^4 - (4ae\cos\alpha)r^3 + r^2(-4ael\cos\alpha + 4a^2\cos^2\alpha - 4a^2(e^2-1)\sin^2\alpha$

$+ (-8a^2\cos^2\alpha - 8a^2l\sin^2\alpha)r + 4a^2l^2\cos^2\alpha + 4a^2l^2\sin^2\alpha$

$= 0$

$\Rightarrow e^2r^4 - (4ae\cos\alpha)r^3 + (4a^2 + 4ael\cos\alpha - 4a^2e^2\sin^2\alpha)r^2$

$- 8a^2lr + 4a^2l^2 = 0$ ——(3)

If r_1, r_2, r_3 and r_4 be the four roots of this eqn (3) then,

$$r_1 r_2 r_3 + r_2 r_3 r_4 + r_1 r_2 r_4 + r_1 r_3 r_4 = + \frac{8a^2 l}{e^2} \quad \text{— (4)}$$

and $$r_1 r_2 r_3 r_4 = \frac{4a^2 l^2}{e^2} \quad \text{— (5)}$$

$$\therefore \quad \frac{r_1 r_2 r_3 + r_2 r_3 r_4 + r_1 r_2 r_4 + r_1 r_3 r_4}{r_1 r_2 r_3 r_4} = \frac{8a^2 l / e^2}{4a^2 l^2 / e^2}$$

$$\Rightarrow \quad \frac{1}{r_1} + \frac{1}{r_2} + \frac{1}{r_3} + \frac{1}{r_4} = \frac{2}{l}$$

2. Show that the equation of the tangent to the conic $\frac{l}{r} = 1 + e\cos\theta$ parallel to the tangent at $\theta = \alpha$ is given by $l(e^2 + 2e\cos\alpha + 1) = p(e^2 - 1)[\cos(\theta - \alpha) + e\cos\theta]$.

Sol^n. Eq^n of tangent at α is

$$\frac{l}{r} = e\cos\theta + \cos(\theta - \alpha). \quad \text{— (1)} \Rightarrow \frac{l}{r} = (e + \cos\alpha)\cos\theta + \sin\alpha\sin\theta$$

Let a tangent be at β.

Eq^n of tangent at β is

$$\frac{l}{r} = e\cos\theta + \cos(\theta - \beta). \quad \text{— (2)} \Rightarrow \frac{l}{r} = (e + \cos\beta)\cos\theta + \sin\beta\sin\theta.$$

(1) & (2) are parallel $\therefore$ $\frac{e + \cos\alpha}{e + \cos\beta} = \frac{\sin\alpha}{\sin\beta} = \frac{1}{K}$ (say)

Then, $\sin\beta = k\sin\alpha$ and, $e+\cos\beta = k(e+\cos\alpha)$

$\Rightarrow \sin\beta = k\sin\alpha$ and $\cos\beta = k(e+\cos\alpha) - e$ ——— (3)

$\sin^2\beta + \cos^2\beta = 1 \Rightarrow k^2\sin^2\alpha + k^2(e+\cos\alpha)^2 + e^2 - 2ke(e+\cos\alpha) = 1$

$\Rightarrow k^2(\sin^2\alpha + e^2 + \cos^2\alpha + 2e\cos\alpha) - 2k(e^2 + e\cos\alpha) + e^2 - 1 = 0$

$\Rightarrow k^2(1 + e^2 + 2e\cos\alpha) - 2ke(e+\cos\alpha) + e^2 - 1 = 0$

~~$\therefore (k-1)$~~ for $k=1$ we have

$1 + e^2 + 2e\cos\alpha - 2e(e+\cos\alpha) + e^2 - 1$

$= 1 + e^2 + 2e\cos\alpha - 2e^2 - 2e\cos\alpha + e^2 - 1$

$= 0 \quad \therefore (k-1)$ is a factor

$\therefore (k-1)\left[k(1+e^2+2e\cos\alpha)\right] + k(1+e^2+2e\cos\alpha) - 2ke(e+\cos\alpha) + e^2 - 1 = 0$

$\Rightarrow (k-1)\left[k(1+e^2+2e\cos\alpha)\right] + (k-1)(k(e^2+1) - 2ke^2 + e^2 - 1 = 0$

$\Rightarrow (k-1)\left[k(1+e^2+2e\cos\alpha)\right] + k - ke^2 + e^2 - 1 = 0$

$\Rightarrow (k-1)\left[k(1+e^2+2e\cos\alpha)\right] + (k-1)(1-e^2) = 0$

$\Rightarrow (k-1)\left[k(1+e^2+2e\cos\alpha) - (e^2-1)\right] = 0$

Since $k \neq 1$ $\therefore k = \dfrac{e^2-1}{1+e^2+2e\cos\alpha}$ ——— (4)

~~$\sin\beta = \dfrac{(e^2-1)\sin\alpha}{\quad}$~~

from (2) $\dfrac{l}{r} = e\cos\theta + \cos\theta\cos\beta + \sin\theta\sin\beta$

using (3) $= e\cos\theta + \cos\theta\left[k(e+\cos\alpha) - e\right] + \sin\theta\, k\sin\alpha$

$= \cancel{e\cos\theta} + \left[e + k(e+\cos\alpha) - e\right]\cos\theta + k\sin\theta\sin\alpha$

$$\Rightarrow \frac{l}{r} = k(e+\cos\alpha)_{1}^{\cos\theta} + k\sin\theta\sin\alpha$$

$$= k\left[e\cos\theta + \cos\theta\cos\alpha + \sin\theta\sin\alpha\right]$$

$$= k\left[e\cos\theta + \cos(\theta-\alpha)\right].$$

From (4)

$$\frac{l}{r} = \frac{e^2-1}{e^2+2e\cos\alpha+1}\left\{e\cos\theta + \cos(\theta-\alpha)\right\}.$$

$$\Rightarrow l\left(e^2+2e\cos\alpha+1\right) = r\left(e^2-1\right)\left\{e\cos\theta + \cos(\theta-\alpha)\right\}.$$

proved.

3. Prove that two conics $\frac{l_1}{r} = 1 - e_1\cos\theta$ and $\frac{l_2}{r} = 1 - e_2\cos(\theta-\alpha)$ will touch each other if $l_1^2(1-e_2^2) + l_2^2(1-e_1^2) = 2l_1l_2(1-e_1e_2\cos\alpha)$.

Sol^n. Let the conics touch at $\theta = \beta$.

Eq^n of tangent at 'β' to $\frac{l_1}{r} = 1 - e_1\cos\theta$ is

$$\frac{l_1}{r} = -e_1\cos\theta + \cos(\theta-\beta). \quad \text{——(1)}$$

and to $\frac{l_2}{r} = 1 - e_2\cos(\theta-\alpha)$ is

$$\frac{l_2}{r} = -e_2\cos(\theta-\alpha) + \cos(\theta-\beta). \quad \text{——(2)}$$

Since (1) and (2) are identical

$$\therefore \frac{l_1}{l_2} = \frac{-e_1+\cos\beta}{-e_2\cos\alpha+\cos\beta} = \frac{\sin\beta}{-e_2\sin\alpha+\sin\beta}$$

i.e $l_1(\cos\beta - e_2\cos\alpha) = l_2(\cos\beta - e_1)$

$\Rightarrow (l_1 - l_2)\cos\beta = l_1 e_2\cos\alpha - e_1 l_2$

and $l_1(\sin\beta - e_2\sin\alpha) = l_2\sin\beta \Rightarrow (l_1 - l_2)\sin\beta = e_2 l_1 \sin\alpha$

Squaring and adding

$$(l_1 - l_2)^2(\cos^2\beta + \sin^2\beta) = (l_1 e_2\cos\alpha - e_1 l_2)^2 + (e_2 l_1 \sin\alpha)^2$$

$$\Rightarrow (l_1 - l_2)^2 = l_1^2 e_2^2\cos^2\alpha + e_1^2 l_2^2 - 2 l_1 l_2 e_1 e_2\cos\alpha + l_1^2 e_2^2\sin^2\alpha$$

$$\Rightarrow l_1^2(1 - e_2^2) + l_2^2(1 - e_1^2) = 2 l_1 l_2(1 - e_1 e_2\cos\alpha).$$

proved.

4. Show that the equation of the circle, which passes through the focus of the parabola $\frac{2a}{r} = 1 + \cos\theta$ and touches it at the point $\theta = \alpha$ is given by $r\cos^3\frac{\alpha}{2} = a\cos(\theta - \frac{3}{2}\alpha)$.

Sol^n. Let the eq^n of the circle ~~be~~ ~~at~~ passing through focus be

$r = 2b\cos(\theta - \beta)$. ——(1) where radius = b & centre is at (b, β).

& circle (1) touches the parabola $\frac{2a}{r} = 1 + \cos\theta$ at $\theta = \alpha$ ——(2)

Eq^n of tangent to parabola (2) at $\theta = \alpha$ is

$\frac{2a}{r} = \cos\theta + \cos(\theta - \alpha)$. ——(3)

Eq^n of normal to (2) at $\theta = \alpha$ is the line $\perp$ to (3)

Let its eq^n be

$$\frac{k}{r} = \cos\left(\frac{\pi}{2}+\theta\right) + \cos\left(\frac{\pi}{2}+\theta-\alpha\right).$$

$$= \sin\theta + \sin(\theta-\alpha). \quad \text{---- (4)}$$

Since (4) passes through $\left(\frac{2a}{1+\cos\alpha}, \alpha\right)$

$$\therefore \frac{k}{2a/(1+\cos\alpha)} = \sin\alpha \Rightarrow k = \frac{2a\sin\alpha}{1+\cos\alpha}.$$

$\therefore$ Eq^{n} of (4) is

$$\frac{2a\sin\alpha}{r(1+\cos\alpha)} = \sin\theta + \sin(\theta-\alpha). \quad \text{---- (5).}$$

Since the circle touches the parabola $\therefore$ the normal to the parabola at $\beta=\alpha$ passes through the centre of the circle i.e (b,β).

$\therefore$ ~~(5) is b~~, (5) passes through (b,β).

$$\frac{2a\sin\alpha}{b(1+\cos\alpha)} = \sin\beta + \sin(\beta-\alpha). \quad \text{---- (6).}$$

More over $\theta=\alpha$ lies on (1) & (2) both

$$\therefore 2b\cos(\alpha-\beta) = \frac{2a}{1+\cos\alpha}. \quad \text{---- (7)}$$

from (6) $b(1+\cos\alpha) = \dfrac{2a\sin\alpha}{\sin\beta+\sin(\beta-\alpha)}$

$\therefore$ (7) $\Rightarrow 2\cos(\alpha-\beta)\cdot\dfrac{2a\sin\alpha}{\sin\beta+\sin(\beta-\alpha)} = 2a.$

$\Rightarrow 2\cos(\alpha-\beta)\sin\alpha = \sin\beta + \sin(\beta-\alpha).$

$\Rightarrow \sin(2\alpha-\beta) + \sin\beta = 2\sin\beta + \sin(\beta-\alpha)$

$\Rightarrow \sin(2\alpha-\beta) = \sin(\beta-\alpha).$

$\Rightarrow 2\alpha-\beta = \beta-\alpha \Rightarrow 3\alpha = 2\beta \Rightarrow \beta = \frac{3\alpha}{2}.$

$\therefore$ from (7) $b = \dfrac{2a\sin\alpha}{\left\{\sin\frac{3\alpha}{2} + \sin\left(\frac{3\alpha}{2}\right)\right\}(1+\cos\alpha)}$

$= \dfrac{2a}{(1+\cos\alpha)\cos\left(\alpha - \right.}$

$\cos\left(\alpha - \frac{3}{2}\alpha\right) = \dfrac{2a}{1+\cos\alpha} \Rightarrow b = \dfrac{a}{(1+\cos\alpha)\cos\frac{\alpha}{2}}$.

$\Rightarrow b = \dfrac{a}{2\cos^3\frac{\alpha}{2}}$

Thus from (1) putting $b = \dfrac{a}{2\cos^3\frac{\alpha}{2}}$ & $\beta = \frac{3}{2}\alpha$ we have

$r = \dfrac{a}{\cos^3\frac{\alpha}{2}}\cos\left(\theta - \frac{3\alpha}{2}\right)$

$\Rightarrow r\cos^3\frac{\alpha}{2} = a\cos\left(\theta - \frac{3}{2}\alpha\right)$. proved.

5. Find the ~~dir~~ equation of the director circle of the conic $\frac{l}{r} = 1 - e\cos\theta$.

Sol^n. Def of Director Circle : Locus of point of intersection of $\perp$ tangents to a conic.

* Let the tangents from (r_1, θ_1) touch the conic $\frac{l}{r} = 1 - e\cos\theta$ at 'α' and 'β'. ——(1)

Eq^n of tangents at 'α' & 'β' to (1) are

$\frac{l}{r} = -e\cos\theta + \cos(\theta - \alpha)$ ——(2)

& $\frac{l}{r} = -e\cos\theta + \cos(\theta - \beta)$ ——(3).

Since they pass through (r_1, θ_1) then

$$\frac{l}{r_1} = -e\cos\theta_1 + \cos(\theta_1 - \alpha)$$

& $$\frac{l}{r_1} = -e\cos\theta_1 + \cos(\theta_1 - \beta)$$

$\Rightarrow \cos(\theta_1 - \alpha) = \cos(\theta_1 - \beta) \Rightarrow \theta_1 - \alpha = \pm(\theta_1 - \beta)$

Since $\theta_1 - \alpha = \theta_1 - \beta$

$\Rightarrow \alpha = \beta$ a contradiction $\therefore \theta_1 - \alpha = -(\theta_1 - \beta)$

$$\Rightarrow \theta_1 = \frac{\alpha+\beta}{2} \quad \text{---- (4)}$$

and $$\frac{l}{r_1} = -e\cos\left(\frac{\alpha+\beta}{2}\right) + \cos\left(\beta - \frac{\alpha}{2}\right) \text{---- (5).}$$

Again since (2) and (3) are ~~be~~ $\perp$ to each other

i.e $$\frac{l}{r} = (-e + \cos\alpha)\cos\theta + \sin\alpha\sin\theta$$

& $$\frac{l}{r} = (-e + \cos\beta)\cos\theta + \sin\beta\sin\theta$$ are $\perp$

$$\therefore (-e + \cos\alpha)(-e + \cos\beta) + \sin\alpha\sin\beta = 0.$$

$$\Rightarrow e^2 - e(\cos\alpha + \cos\beta) + \cos\alpha\cos\beta + \sin\alpha\sin\beta = 0$$

$$\Rightarrow e^2 - 2e\cos\left(\frac{\alpha+\beta}{2}\right)\cos\left(\frac{\alpha-\beta}{2}\right) + \cos(\alpha-\beta) = 0$$

$$\Rightarrow e^2 - 2e\cos\left(\frac{\alpha+\beta}{2}\right)\cos\left(\frac{\alpha-\beta}{2}\right) + 2\cos^2\left(\frac{\alpha-\beta}{2}\right) - 1 = 0.$$

from (4)

$$e^2 - 2e\cos\theta_1\cos\left(\frac{\alpha-\beta}{2}\right) + 2\cos^2\left(\frac{\alpha-\beta}{2}\right) - 1 = 0 \quad \text{---- (6)}$$

from (5) $$\frac{l}{r_1} = -e\cos\theta_1 + \cos\left(\beta - \frac{\alpha}{2}\right)$$

$$\Rightarrow \cos\left(\frac{\alpha-\beta}{2}\right) = \frac{l}{r_1} + e\cos\theta_1$$

$\therefore$ (6) becomes

$$e^2 - 2e\cos\theta_1\left(\frac{l}{r_1} + e\cos\theta_1\right) + 2\left(\frac{l}{r_1} + e\cos\theta_1\right)^2 - 1 = 0$$

$$\Rightarrow e^2 - \frac{2el}{r_1}\cos\theta_1 - 2e^2\cos^2\theta_1 + 2\cdot\frac{l^2}{r_1^2} + 2e^2\cos^2\theta_1 + 4\frac{el}{r_1}\cos\theta_1 - 1 = 0.$$

$\Rightarrow r_1^2(e^2-1) + 2l^2 + 2el r_1 \cos\theta_1 = 0.$

Thus locus of (r_1, θ_1) i.e eqn of director circle is

$$r^2(e^2-1) + 2el r\cos\theta + 2l^2 = 0.$$

6. Find the chord of contact of tangents of the conic $\frac{l}{r} = 1 + e\cos\theta$ from a given point (r_1, θ_1). Hence find the chord of contact of tangents drawn from $(l, 0)$.

Sol^n. Let the tangents from (r_1, θ_1) touch the conic

$$\frac{l}{r} = 1 + e\cos\theta \quad \text{——(1)} \quad \text{at } '\alpha-\beta' \text{ and } '\alpha+\beta'.$$

Eqn of chord joining $'\alpha-\beta'$ & $'\alpha+\beta'$ is

$$\frac{l}{r} = e\cos\theta + \sec\beta\cos(\theta-\alpha) \quad \text{——(2)}.$$

Eqn of tangent at $'\alpha-\beta'$ & $'\alpha+\beta'$ to (1) is

$$\frac{l}{r} = e\cos\theta + \cos(\theta-\alpha+\beta) \quad \text{——(3)}$$

& $$\frac{l}{r} = e\cos\theta + \cos(\theta-\alpha-\beta) \quad \text{——(4)}$$

Both pass through (r_1, θ_1).

$\therefore \frac{l}{r_1} = e\cos\theta_1 + \cos(\theta_1-\alpha+\beta)$

& $\frac{l}{r_1} = e\cos\theta_1 + \cos(\theta_1-\alpha-\beta).$

$\Rightarrow \cos(\theta_1-\alpha+\beta) = \cos(\theta_1-\alpha-\beta) \Rightarrow \theta_1-\alpha+\beta = \pm(\theta_1-\alpha-\beta).$

+ve sign is inadmissible because in that case $\beta = 0$ which is not possible

$\therefore \theta_1-\alpha+\beta = -\theta_1+\alpha+\beta \Rightarrow \alpha = \theta_1 \quad \text{——(5)}$

Again $\frac{l}{r_1} = e\cos\theta_1 + \cos\beta \Rightarrow \cos\beta = \frac{l}{r_1} - e\cos\theta_1 \quad \text{——(6)}$

Thus from (2) we have

$$\frac{l}{r} = e\cos\theta + \frac{\cos(\theta-\theta_1)}{\frac{l}{r_1} - e\cos\theta_1} \Rightarrow \left(\frac{l}{r} - e\cos\theta\right)\left(\frac{l}{r_1} - e\cos\theta_1\right) = \cos(\theta-\theta_1).$$

※ Show that the polar equation of the st. line passing through the point (r_1, θ_1) and (r_2, θ_2) is

$$\frac{1}{r}\sin(\theta_1 - \theta_2) - \frac{1}{r_1}\sin(\theta - \theta_2) + \frac{1}{r_2}\sin(\theta - \theta_1) = 0$$

Hence find the condition of collinearity of the points (r_i, θ_i) $i = 1, 2, 3$.

Sol^n. Let the eq^n of the st. line be

$$\frac{1}{r} = A\cos\theta + B\sin\theta. \quad \text{——— (1)}$$

Since (r_1, θ_1) & (r_2, θ_2) pass through it

$$\therefore \frac{1}{r_1} = A\cos\theta_1 + B\sin\theta_1 \; ; \; \frac{1}{r_2} = A\cos\theta_2 + B\sin\theta_2$$

i.e $$\frac{1}{r_1}\cos\theta_2 - \frac{1}{r_2}\cos\theta_1 = B\sin(\theta_1 - \theta_2).$$

and $$\frac{1}{r_1}\sin\theta_2 - \frac{1}{r_2}\sin\theta_1 = A\sin(\theta_2 - \theta_1).$$

Thus (1) becomes $$\frac{1}{r} = \frac{1}{\sin(\theta_2 - \theta_1)}\left\{\frac{1}{r_1}\sin\theta_2 - \frac{1}{r_2}\sin\theta_1\right\}\cos\theta$$

$$+ \frac{1}{\sin(\theta_1 - \theta_2)}\left\{\frac{1}{r_1}\cos\theta_2 - \frac{1}{r_2}\cos\theta_1\right\}\sin\theta$$

$$\Rightarrow \frac{\sin(\theta_1 - \theta_2)}{r} = \frac{1}{r_1}(\cos\theta_2\sin\theta - \sin\theta_2\cos\theta) + \frac{1}{r_2}(\cos\theta_1\sin\theta - \sin\theta_1\cos\theta)$$

$$\Rightarrow \frac{\sin(\theta_1 - \theta_2)}{r} - \frac{1}{r_1}\sin(\theta - \theta_2) + \frac{1}{r_2}\sin(\theta - \theta_1) = 0. \text{ proved.}$$

——— (2)

Cond. of (r_3, θ_3) will lie on (2).
collinearity

$$\therefore \frac{\sin(\theta_1 - \theta_2)}{r_3} + \frac{\sin(\theta_3 - \theta_2)}{r_1} + \frac{\sin(\theta_3 - \theta_1)}{r_2} = 0.$$

1. A variable plane cuts the axes in A, B, C and the centroid of the triangle ABC is (a, b, c). Show that the equation of the plane is $\frac{x}{a} + \frac{y}{b} + \frac{z}{c} = 3$.

Sol^n. Let the co-ordinates of A, B and C be respectively $(\alpha, 0, 0)$, $(0, \beta, 0)$ and $(0, 0, \gamma)$.

Then $\frac{\alpha}{3} = a$, $\frac{\beta}{3} = b$ & $\frac{\gamma}{3} = c$ ——(1)

Eq^n of plane cutting intercepts α, β & γ on the x, y & z axis respectively is

$$\frac{x}{\alpha} + \frac{y}{\beta} + \frac{z}{\gamma} = 1.$$

$$\Rightarrow \frac{x}{3a} + \frac{y}{3b} + \frac{z}{3c} = 1 \Rightarrow \frac{x}{a} + \frac{y}{b} + \frac{z}{c} = 3.$$

19(i)

2. A variable plane which is at a constant distance 3p from the origin O cuts the axes in A, B, C. Show that the locus of the centroid of the triangle ABC is $\frac{1}{x^2} + \frac{1}{y^2} + \frac{1}{z^2} = \frac{1}{p^2}$

Sol^n. Let the co-ordinates of A, B, C be resp. $(\alpha, 0, 0)$, $(0, \beta, 0)$ and $(0, 0, \gamma)$.

Eq^n of plane through A, B, C is $\frac{x}{\alpha} + \frac{y}{\beta} + \frac{z}{\gamma} = 1$. ——(1).

The centroid of the triangle ABC is $\left(\frac{\alpha}{3}, \frac{\beta}{3}, \frac{\gamma}{3}\right)$.

Since 3p is the distance of the origin from the plane (1)

then $$\left| \frac{-1}{\sqrt{\left(\frac{1}{\alpha}\right)^2 + \left(\frac{1}{\beta}\right)^2 + \left(\frac{1}{\gamma}\right)^2}} \right| = 3p. \Rightarrow \frac{1}{\frac{1}{\alpha^2} + \frac{1}{\beta^2} + \frac{1}{\gamma^2}} = 9p^2$$

$$\Rightarrow \frac{1}{9p^2} = \frac{1}{\alpha^2} + \frac{1}{\beta^2} + \frac{1}{\gamma^2} \Rightarrow \frac{1}{p^2} = \left(\frac{3}{\alpha}\right)^2 + \left(\frac{3}{\beta}\right)^2 + \left(\frac{3}{\gamma}\right)^2.$$

Since the centroid is $\left(\frac{\alpha}{3}, \frac{\beta}{3}, \frac{\gamma}{3}\right)$ hence the locus of the centroid is

$$\frac{1}{p^2} = \frac{1}{x^2} + \frac{1}{y^2} + \frac{1}{z^2}.$$

3[21]. A variable plane has intercepts on the co-ord. axes the sum of whose squares is a constant k^2. Show that the locus of the foot of the perpendicular from the origin to the plane is

$$(x^2 + y^2 + z^2)^2 (x^{-2} + y^{-2} + z^{-2}) = k^2$$

Sol[n].

Let the plane intercept lengths α, β, γ on the x, y & z axes resp.

Then $\alpha^2 + \beta^2 + \gamma^2 = k^2$ —— (1).

Eq[n] of plane through $(\alpha, 0, 0)$, $(0, \beta, 0)$ and $(0, 0, \gamma)$ is

$$\frac{x}{\alpha} + \frac{y}{\beta} + \frac{z}{\gamma} = 1 \text{ ——— (2)}.$$

If the foot of the ⊥ from O(0, 0, 0) to the plane (2) is $N(x_1, y_1, z_1)$ then the ~~d.r. of~~ normal to the plane and ON are ||.

$$\therefore \frac{\frac{1}{\alpha}}{x_1} = \frac{\frac{1}{\beta}}{y_1} = \frac{\frac{1}{\gamma}}{z_1} = \lambda \text{ (say)}.$$

$$\Rightarrow \frac{1}{\alpha} = \lambda x_1, \ \frac{1}{\beta} = \lambda y_1, \ \frac{1}{\gamma} = \lambda z_1. \text{ ——— (3)}.$$

Since N lies on (2) $\therefore \frac{x_1}{\alpha} + \frac{y_1}{\beta} + \frac{z_1}{\gamma} = 1$

$$\Rightarrow \lambda x_1^2 + \lambda y_1^2 + \lambda z_1^2 = 1 \text{ from (3)} \Rightarrow \lambda = \frac{1}{x_1^2 + y_1^2 + z_1^2} \text{ ——— (4)}$$

$$\frac{x+3}{} = \frac{y-5}{} = \frac{z-2}{}$$

from (1) and (3) we eliminate α, β, γ to get

$$\frac{1}{\lambda^2 x_1^2} + \frac{1}{\lambda^2 y_1^2} + \frac{1}{\lambda^2 z_1^2} = k^2$$

$$\Rightarrow \quad x_1^{-2} + y_1^{-2} + z_1^{-2} = k^2 \cdot \lambda^2$$

$$= k^2 \left(\frac{1}{x_1^2 + y_1^2 + z_1^2}\right)^2 \cdot \text{from (4)}.$$

$$\Rightarrow \left(x_1^2 + y_1^2 + z_1^2\right)^2 \left(x_1^{-2} + y_1^{-2} + z_1^{-2}\right) = k^2$$

Thus the locus of (x_1, y_1, z_1) is

$$\left(x^2 + y^2 + z^2\right)^2 \left(x^{-2} + y^{-2} + z^{-2}\right) = k^2.$$

4. A variable plane passes through the point (f, g, h) and meets the axes in A, B, C. If the planes through A, B, C and $\parallel$ to the co-ord planes meet at P, show that the locus of P is

$$f x^{-1} + g y^{-1} + h z^{-1} = 1.$$

Sol^n. Let $A \equiv (\alpha, 0, 0)$ $B \equiv (0, \beta, 0)$, $C \equiv (0, 0, \gamma)$.

Eq^n of plane through A and $\parallel$ to yz plane is $1 \cdot (x - \alpha) = 0$

$$\Rightarrow x = \alpha. \quad \text{---(1)}$$

" " " " B " " " xz plane is $y = \beta$ ---(2)

C " " xy " " $z = \gamma$ ---(3)

Point of intersection of (1), (2) and (3) is $P \equiv (\alpha, \beta, \gamma)$.

Eq^n of plane through A, B, C is $\frac{x}{\alpha} + \frac{y}{\beta} + \frac{z}{\gamma} = 1$ ---(4)

It passes through (f, g, h). Then $\frac{f}{\alpha} + \frac{g}{\beta} + \frac{h}{\gamma} = 1$ ---(5)

Thus locus of P is $f x^{-1} + g y^{-1} + h z^{-1} = 1$.

A point P moves on the plane $\frac{x}{a} + \frac{y}{b} + \frac{z}{c} = 1$ which is fixed and the plane through P ⊥ to OP meets the axes in A, B, C. If the planes through A, B, C and ∥ to the co-ord planes meet in a point Q then show that the locus of Q is $k^2x + k^2y + k^2z = \frac{1}{a^2} + \frac{1}{b^2} + \frac{1}{c^2}$.

Let $P \equiv (\alpha, \beta, \gamma)$.

Eqn of plane ⊥ to OP and passing through P is

$$\alpha(x - \alpha) + \beta(y - \beta) + \gamma(z - \gamma) = 0 \quad \text{——(1)}$$

$$\Rightarrow x\alpha + y\beta + z\gamma = \alpha^2 + \beta^2 + \gamma^2$$

$$\Rightarrow \frac{x}{\frac{\alpha^2+\beta^2+\gamma^2}{\alpha}} + \frac{y}{\frac{\alpha^2+\beta^2+\gamma^2}{\beta}} + \frac{z}{\frac{\alpha^2+\beta^2+\gamma^2}{\gamma}} = 1$$

Then co-ord of $A \equiv \left(\frac{\alpha^2+\beta^2+\gamma^2}{\alpha}, 0, 0\right) \equiv \left(\frac{k}{\alpha}, 0, 0\right)$

$$B \equiv \left(0, \frac{\alpha^2+\beta^2+\gamma^2}{\beta}, 0\right) \equiv \left(0, \frac{k}{\beta}, 0\right)$$

$$C \equiv \left(0, 0, \frac{\alpha^2+\beta^2+\gamma^2}{\gamma}\right) \equiv \left(0, 0, \frac{k}{\gamma}\right)$$

where $k = \alpha^2 + \beta^2 + \gamma^2$

Eqn of plane passing through A and ∥ to yz plane

$$\text{is} \quad (x - \frac{k}{\alpha}) = 0 \Rightarrow x = \frac{k}{\alpha}$$

Eqn of plane ... B ∥ to xz plane

$$\text{is} \quad (y - \frac{k}{\beta}) = 0 \Rightarrow y = \frac{k}{\beta}$$

Eqn of plane ... C ... yz plane is $z = \frac{k}{\gamma}$

These planes meet at $Q \equiv \left(\frac{k}{\alpha}, \frac{k}{\beta}, \frac{k}{\gamma}\right)$.

Since P lies on $\frac{x}{a} + \frac{y}{b} + \frac{z}{c} = 1$ then

$$\frac{\alpha}{a} + \frac{\beta}{b} + \frac{\gamma}{c} = 1$$

$$\Rightarrow \frac{\alpha}{ka} + \frac{\beta}{kb} + \frac{\gamma}{kc} = \frac{1}{k} \quad \text{——— (1)}.$$

Let- $Q \equiv (\tilde{x}, \tilde{y}, \tilde{z})$. then $\tilde{x} = \frac{k}{\alpha}$

$\tilde{y} = \frac{k}{\beta}$

$\tilde{z} = \frac{k}{\gamma}$.

$$\Rightarrow \frac{\alpha^2}{x} + \frac{\beta^2}{y} + \frac{\gamma^2}{z} = k^2\left(\frac{1}{\alpha^2} + \frac{1}{\beta^2} + \frac{1}{\gamma^2}\right).$$

$$\Rightarrow \left(\frac{1}{\tilde{x}}\right)^2 + \left(\frac{1}{\tilde{y}}\right)^2 + \left(\frac{1}{\tilde{z}}\right)^2 = \frac{\alpha^2+\beta^2+\gamma^2}{k^2}$$

$$= \frac{\alpha^2+\beta^2+\gamma^2}{(\alpha^2+\beta^2+\gamma^2)^2}$$

$$= \frac{1}{\alpha^2+\beta^2+\gamma^2}.$$

$$\Rightarrow \alpha^2+\beta^2+\gamma^2 = k$$

$$\Rightarrow \frac{1}{k} = \frac{1}{\tilde{x}^2} + \frac{1}{\tilde{y}^2} + \frac{1}{\tilde{z}^2}$$

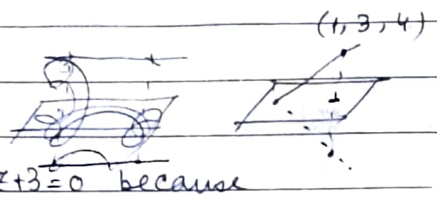
from (1). $\frac{1}{\tilde{x}a} + \frac{1}{\tilde{y}b} + \frac{1}{\tilde{z}c} = \frac{1}{\tilde{x}^2} + \frac{1}{\tilde{y}^2} + \frac{1}{\tilde{z}^2}$.

Thus locus of $(\tilde{x}, \tilde{y}, \tilde{z})$ is

$$\frac{1}{ax} + \frac{1}{by} + \frac{1}{cz} = \frac{1}{x^2} + \frac{1}{y^2} + \frac{1}{z^2}.$$

Find the image of the st. line $\frac{x-1}{3} = \frac{y-3}{5} = \frac{z-4}{2}$ in the plane $2x - y + z + 3 = 0$.

Sol: The st. line $\frac{x-1}{3} = \frac{y-3}{5} = \frac{z-4}{2}$



is not ∥ to the plane $2x - y + z + 3 = 0$ because

$2 \times 2 + 5 \times (-1) + 2 \times 3 = 6 - 5 + 6 \neq 0$.

Image of $P(1, 3, 4)$ w.r.t. on the plane be P'.

Eq^n of PP' is $\frac{x-1}{2} = \frac{y-3}{-1} = \frac{z-4}{1}$

$P' \equiv (2\lambda + 1, -\lambda + 3, \lambda + 4)$.

The mid pt. of PP' is $\left(\frac{1 + 2\lambda + 1}{2}, \frac{3 - \lambda + 3}{2}, \frac{4 + \lambda + 4}{2}\right)$

The mid pt. lies on the plane.

$\therefore 2\left[\frac{2(\lambda+1)}{2}\right] - \left[\frac{6-\lambda}{2}\right] + \left[\frac{8+\lambda}{2}\right] + 3 = 0$.

$\Rightarrow 2(\lambda+1) - 3 + \frac{\lambda}{2} + 4 + \frac{\lambda}{2} + 3 = 0$.

$\Rightarrow 2\lambda + 2 - 3 + \lambda + 4 + 3 = 0 \Rightarrow 3\lambda + 6 = 0$

$\lambda = -2$

$\therefore P' \equiv (-4+1, 2+3, -3+4) = (-3, 5, 2)$.

Again intersection of the st. line with the plane be Q.

$Q : (3r + 1, 5r + 3, 2r + 4)$

Q lies on the plane.

$\therefore 2(3r+1) - (5r+3) + (2r+4) + 3 = 0$.

$\Rightarrow 6r - 5r + 2r + 2 - 3 + 4 + 3 = 0$

$\Rightarrow 3r + 6 = 0 \Rightarrow r = -2$.

$\therefore Q \equiv (-6+1, -10+3, -4+4) = (-5, -7, 0)$.

Hence eq^n of QP' is

$$\frac{x+3}{2} = \frac{y-5}{12} = \frac{z-2}{2}$$

$$= 1 \qquad = 6 \qquad = 1$$

Show that the dist. of the pt. $(3, 8, 2)$ from the st. line ... the point $(3, 8, 2)$ from the st. line

Show that the dist. of the pt. $(3, -4, 5)$ from the plane $2x + 5y - 6z = 0$ along the straight line $\frac{x-3}{2} = \frac{y+4}{-1} = \frac{z-5}{-2}$ is 12 units.

Soln Eqn of st. line passing through $P(3, -4, 5)$ ~~along~~ $\parallel$ to the st. line whose direction ratios are $(2, -1, -2)$ is

$$\frac{x-3}{2} = \frac{y+4}{-1} = \frac{z-5}{-2} \quad \text{——— (1)}$$

~~Any pt on this~~ (+) Pt. of intersection of (1) with the plane $2x + 5y - 6z = 0$ ——— (2) is be Q.

Co-ord. of $Q \equiv (2\lambda + 3, -\lambda - 4, -2\lambda + 5)$

Q lies on (2)

$\therefore 2(2\lambda + 3) + 5(-\lambda - 4) - 6(-2\lambda + 5) = 0.$

$\Rightarrow 4\lambda - 5\lambda + 12\lambda + 6 - 20 - 30 = 0.$

$\Rightarrow 11\lambda - 44 = 0 \Rightarrow \lambda = 4.$

$\therefore Q \equiv (11, -8, -3).$

$\therefore PQ^2 = (3-11)^2 + (-4+8)^2 + (5+3)^2$

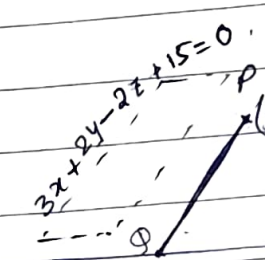
$= 64 + 16 + 64$

$= 144$

$\therefore PQ = 12$ units.

Show that the distance of the point $(3, 8, 2)$ from the st. line $\frac{x-1}{2} = \frac{y-3}{4} = \frac{z-2}{3}$ along the plane $3x + 2y - 2z + 15 = 0$ is 7 units.

Sol^n Eq^n of st. line ∥ to the plane:



$$\frac{x-1}{2} = \frac{y-3}{4} = \frac{z-2}{3} \quad \text{—(1)}$$

Let the st. line through $P(3, 8, 2)$, ∥ to the plane meet the st. line (1) at Q.

Let $Q \equiv (2\lambda+1, 4\lambda+3, 3\lambda+2)$.

D.r. of PQ is $(2\lambda-2, 4\lambda-5, 3\lambda)$

PQ is ⊥ to the normal to the plane.

$\therefore 3(2\lambda-2) + 2(4\lambda-5) - 2(3\lambda) = 0.$

$\Rightarrow 6\lambda + 8\lambda - 6\lambda - 6 - 10 = 0 \Rightarrow 8\lambda = 16$

$\Rightarrow \lambda = 2.$

$\therefore Q \equiv (5, 11, 8).$

$\therefore PQ^2 = 2^2 + 3^2 + 6^2$

$= 4 + 9 + 36$

$= 49 \Rightarrow PQ = 7$ units.

CONE

Surface generated by a straight line which always passes through a fixed point (vertex) and intersecting a given curve (guiding curve). The moving straight line is called the generator.

(2) I. Cone with vertex at the origin.

Let general eqn of second degree:

$$ax^2 + by^2 + cz^2 + 2hxy + 2fyz + 2gzx + 2ux + 2vy + 2wz + d = 0 \quad \text{——(1)}$$

represents a cone passing through the origin.

Let $P(x_1, y_1, z_1)$ be a point on the cone. Then OP is a generator & every pt. on OP must lie on (1).

eqn. of OP : $\frac{x}{x_1} = \frac{y}{y_1} = \frac{z}{z_1}$.

any pt. on OP : (rx_1, ry_1, rz_1)

Thus every pt. on OP must satify (1).

i.e. $a(rx_1)^2 + \ldots\ldots + d = 0$

i.e $r^2(ax_1^2 + by_1^2 + cz_1^2 + 2fy_1z_1 + 2hx_1y_1 + 2gx_1z_1)$
$+ 2r(ux_1 + vy_1 + wz_1) + d = 0$. must hold ~~for~~ $\forall$ r.

i.e $ax_1^2 + by_1^2 + cz_1^2 + 2fy_1z_1 + 2hx_1y_1 + 2gx_1z_1 = 0$

$ux_1 + vy_1 + wz_1 = 0 \qquad d = 0$.

Since (x_1, y_1, z_1) lies on a cone $\therefore ux_1 + vy_1 + wz_1 = 0$ holds only when $u = 0 = v = w$.

Thus (1) becomes $ax^2 + by^2 + cz^2 + 2hxy + 2fyz + 2gzx = 0$.

Converse is also true.

(4) Condition for general eq^n of 2nd degree to represent a cone.

Let $ax^2+by^2+cz^2+2hxy+2fyz+2gzx+2ux+2vy+2wz+d=0$ ——(1)
represent a cone with vertex at (α,β,γ).

Let the origin be shifted to (α,β,γ). Then (1) becomes

$$a(x+\alpha)^2+b(y+\beta)^2+\ldots\ldots+d=0.$$

$$\Rightarrow ax^2+by^2+cz^2+2fyz+2hxy+2gzx$$
$$+2x(a\alpha+h\beta+g\gamma+u)+2y(h\alpha+b\beta+f\gamma+v)$$
$$+2z(g\alpha+f\beta+c\gamma+w)+a\alpha^2+b\beta^2+c\gamma^2+2f\beta\gamma$$
$$+2g\gamma\alpha+2h\alpha\beta+2u\alpha+2v\beta+2w\gamma+d=0.$$

This equation represents a cone with vertex at origin.

$$\therefore\quad a\alpha+h\beta+g\gamma+u=0 \quad\text{——(2)}$$
$$h\alpha+b\beta+f\gamma+v=0 \quad\text{——(3)}$$
$$g\alpha+f\beta+c\gamma+w=0 \quad\text{——(4)}$$
$$a\alpha^2+b\beta^2+c\gamma^2+2f\beta\gamma+2h\alpha\beta+2g\alpha\gamma+2u\alpha+2v\beta+2w\gamma+d=0 \quad\text{——(5)}$$

$$(5)\rightarrow \alpha(a\alpha+h\beta+g\gamma+u)+\beta(b\beta+h\alpha+f\gamma+v)$$
$$+\gamma(c\gamma+f\beta+g\alpha+w)+u\alpha+v\beta+w\gamma+d=0$$

from (2, 3, 4)
$$\Rightarrow\quad u\alpha+v\beta+w\gamma+d=0 \quad\text{——(6)}.$$

eliminating α,β,γ from 2, 3, 4, 6 we have

$$\begin{vmatrix} a & h & g & u \\ h & b & f & v \\ g & f & c & w \\ u & v & w & d \end{vmatrix}=0$$

→ condition for (1) to represent a cone.

Date
Page

3. (3) Eqⁿ of cone containing the axes.

The general eqⁿ of cone with vertex as origin:

$$ax^2 + by^2 + cz^2 + 2hxy + 2fyz + 2gzx = 0. \quad \text{—(1)}$$

Let the st line $\frac{x}{l} = \frac{y}{m} = \frac{z}{n}$ lie on it.

then, $al^2 + bm^2 + cn^2 + 2hlm + 2fmn + 2gln = 0$. —(2)

If the st. line be the ~~rect~~ x-axis, y & z axis then (l, m, n) ≡ (1, 0, 0), (0, 1, 0) & (0, 0, 1).

i.e from (2) $a = 0, \quad b = 0, \quad c = 0$

from (1) the eqⁿ of cone is

$$hxy + fyz + gzx = 0.$$

(1) 4. Eq. of cone with origin as vertex & a given curve as base.

Let guiding curve be $f(x, y) = 0, z = c$ and the straight line $\frac{x}{l} = \frac{y}{m} = \frac{z}{n}$ be a generator.

Then the st. line meets the plane $z = c$.

i.e $\frac{x}{l} = \frac{y}{m} = \frac{c}{n}$. The pt. of intersection of the st. line and ~~the~~ $z = c$ is $\left(\frac{lc}{n}, \frac{mc}{n}, c\right)$.

This pt. lies on the curve $f(x, y) = 0, z = c$.

$\therefore f\left(\frac{lc}{n}, \frac{mc}{n}\right) = 0.$

or, $f\left(\frac{xc}{z}, \frac{yc}{z}\right) = 0$ [by eliminating $\frac{l}{n}$ & $\frac{m}{n}$].

~~Thus~~ ↳ required eqⁿ of the cone.

Note. If the vertex is at (α, β, γ). the eqⁿ of the line is

$$\frac{x-\alpha}{l} = \frac{y-\beta}{m} = \frac{z-\gamma}{n}.$$

Other things remaining the same.

5. Eqⁿ of a right circular cone.

a. vertex at origin.

axis : $\frac{x}{l} = \frac{y}{m} = \frac{z}{n}$ — (1) semi vertical angle = θ.

Let a, b, c be the d.r. of a generator.

Then generator : $\frac{x}{a} = \frac{y}{b} = \frac{z}{c}$ — (2)

If (2) makes angle θ with (1) then

$$\cos\theta = \frac{la + mb + nc}{\sqrt{l^2+m^2+n^2}\sqrt{a^2+b^2+c^2}}.$$

or, $(l^2+m^2+n^2)(a^2+b^2+c^2)\cos^2\theta = (al + bm + cn)^2$

or, $(l^2+m^2+n^2)(x^2+y^2+z^2)\cos^2\theta = (lx + my + nz)^2$.

b. vertex is at $(\alpha, \beta, \gamma) \neq (0,0,0)$.

axis : $\frac{x-\alpha}{l} = \frac{y-\beta}{m} = \frac{z-\gamma}{n}$.

generator : $\frac{x-\alpha}{a} = \frac{y-\beta}{b} = \frac{z-\gamma}{c}$.

$$\therefore (l^2+m^2+n^2)\left[(x-\alpha)^2 + (y-\beta)^2 + (z-\gamma)^2\right]\cos^2\theta = \left[l(x-\alpha) + m(y-\beta) + n(z-\gamma)\right]^2.$$

6. Intersection of a cone by a plane through vertex.

eqⁿ of cone ~~through~~ having vertex at (0,0) :

$$ax^2 + by^2 + cz^2 + 2fyz + 2gzx + 2hxy = 0.$$

& eqⁿ of plane $ux + vy + wz = 0$.

CYLINDER

A surface generated by a st. line passing (or touching) a given fixed curve (guiding curve) and parallel to a given fixed line line (axis). The moving st. line is a generator.

Right circular Cylinder : If the guiding curve is a circle and the axis is $\perp$ to the plane of the circle passing through the center.

Eqⁿ of a cylinder. guiding curve : $f(x, y) = 0 \; ; z = 0$.

axis : $\frac{x}{l} = \frac{y}{m} = \frac{z}{n}$.

Then eqⁿ of generator : $\frac{x - x'}{l} = \frac{y - y'}{m} = \frac{z - z'}{n}$

where (x', y', z') is any point on the cylinder.

any pt. on the generator is $(lr + x', \; mr + y', \; nr + z')$.

This pt. lies on the guiding curve if

$nr + z' = 0 \Rightarrow r = -z'/n$.

& $f(lr + x', \; mr + y') = 0$

or $f\left(-\frac{lz'}{n} + x', \; -\frac{mz'}{n} + y'\right) = 0$.

The locus of (x', y', z') is $f\left(x - \frac{lz}{n}, \; y - \frac{mz}{n}\right) = 0$ which is the equation of the cylinder.

Eqⁿ of a Right Circular Cylinder :

Let the axis be $\frac{x - \alpha}{l} = \frac{y - \beta}{m} = \frac{z - \gamma}{n}$ & ~~xd~~ radius ~~by~~ be r.

Any point on the cylinder be $P.(x', y', z')$.

Distance of (x', y', z') from the axis $= r$.

i.e.

$$(x'-\alpha)^2 + (y'-\beta)^2 + (z'-\gamma)^2 - \left\{ \frac{l(x'-\alpha) + m(y'-\beta) + n(z'-\gamma)}{\sqrt{l^2+m^2+n^2}} \right\}^2$$

$$= r^2.$$

locus of (x', y', z') is the eqn of the cylinder.

VI B

16. any pt. on the cylinder (x', y', z').

eqn of generator through (x', y', z') is

$$\frac{x-x'}{3} = \frac{y-y'}{5} = \frac{z-z'}{-4}.$$

This intersects $3x^2 - 4y^2 = 5$, $z = 2$ at-

$(3r+x', 5r+y', -4r+z')$.

then, $-4r + z' = 2 \Rightarrow 4r = z' - 2 \Rightarrow r = \frac{z'-2}{4}$.

$$\therefore \quad 3r + x' = \frac{3}{4}(z'-2) + x'$$

$$5r + y' = \frac{5}{4}(z'-2) + y'.$$

$$\therefore 3\left\{\frac{3}{4}(z'-2) + x'\right\}^2 - 4\left\{\frac{5}{4}(z'-2) + y'\right\}^2 = 5.$$

14. $\perp$ from $(3, -1, 1)$ to $\frac{x-1}{2} = \frac{y+3}{-1} = \frac{z-2}{1}$ is the radi

$$r^2 = (3-1)^2 + (-1+3)^2 + (1-2)^2 - \left\{ \frac{2(2) - 1(+2) + 1(-1)}{\sqrt{2^2+1^2+1^2}} \right\}^2$$

$$= 4 + 4 + 1 - \tfrac{1}{6} = 9.53/6$$

Let any pt. be (x', y', z'). on the cylinder. Then

$$(x'-1)^2 + (y'+3)^2 + (z'-2)^2 - \left\{ \frac{2(x'-1) - (y'+3) + (z'-2)}{\sqrt{6}} \right\}^2$$

$$\Rightarrow x'^2 + y'^2 + z'^2 - 2x' + 6y' - 4z' + 14 - \frac{1}{6}(2x' - y' + z' - 7)^2 = \frac{53}{6}$$

$$\Rightarrow 6(x'^2 + y'^2 + z'^2 - 2x' + 6y' - 4z' + 14) - (4x'^2 + y'^2 + z'^2 + 49 - 4x'y' - 2y'z' - 14z' + 4x'z' - 28x' - 14y') = 53$$

$\Rightarrow 2x'^2 + 5y'^2 + 5z'^2 + 16x'y' + 22y' - 10z' + 4x'y' + 2y'z' - 4x'z' = 18$

13. guiding curve : $x^2 + y^2 + z^2 = 4$, $x + y + z = 4$.

axis : $\frac{x}{0} = \frac{y}{1} = \frac{z}{0}$.

Let (α, β, γ) be any pt. on the cylinder.

St. line through (α, β, γ) & ∥ to the axis is

$$\frac{x-\alpha}{0} = \frac{y-\beta}{1} = \frac{z-\gamma}{0}.$$

Any pt. on the st. line : $(\alpha, \kappa+\beta, \gamma)$.

if this point lies on the guiding curve. Then

$\alpha + \kappa + \beta + \gamma = 4 \Rightarrow \kappa = 4 - \alpha - \beta - \gamma$ & $\gamma + \beta = 4 - \alpha - \gamma$.

and, $\alpha^2 + (4 - \alpha - \gamma)^2 + \gamma^2 = 4$.

∴ The locus of (α, β, γ) is the eqⁿ of the cylinder.

i.e. $x^2 + 16 + x^2 + z^2 - 8x - 8z + 2xz + z^2 = 4$

$\Rightarrow 2x^2 + 2z^2 - 8x - 8z + 2xz + 12 = 0$

or, $x^2 + z^2 + xz - 4x - 4z + 6 = 0$.

9. axis : $\frac{x}{l} = \frac{y}{m} = \frac{z}{n}$. —(1)

guiding curve : $ax^2 + by^2 = 1$, $z = 0$. —(2)

generator : $\frac{x-\alpha}{l} = \frac{y-\beta}{m} = \frac{z-\gamma}{n}$. —(3)

any pt on (3) $(lr + \alpha, mr + \beta, nr + \gamma)$. lies on (2).

∴ $nr + \gamma = 0 \Rightarrow r = -\gamma/n$.

& $a\left(-\frac{l\gamma}{n} + \alpha\right)^2 + b\left(-\frac{m\gamma}{n} + \beta\right)^2 = 1$.

locus of (α, β, γ) is

$a(x - lz)^2 + b(y - mz)^2 = n^2$.

8. axis : $2x = y = 3z$: $\frac{x}{1/2} = \frac{y}{1} = \frac{z}{1/3}$ —(1)

generator : $\frac{x-\alpha}{1/2} = \frac{y-\beta}{1} = \frac{z-\gamma}{1/3}$ —(2)

guiding curve : $y = 0$, $x^2 + z^2 = 6$. —(3)

any point on (2) is $\left(\frac{1}{2}r + \alpha, r + \beta, \frac{1}{3}r + \gamma\right)$.

If this point lies on (3) then

$$r + \beta = 0 \rightarrow r = -\beta \quad \text{and} \quad \left(\frac{1}{2}r + \alpha\right)^2 + \left(\frac{1}{3}r + \gamma\right)^2 = 6$$

$$\Rightarrow 9(-\beta + 2\alpha)^2 + 4(-\beta + 3\gamma)^2 = 9 \times 4 \times 6.$$

∴ locus of (α, β, γ) is the equation of the cylinder given as

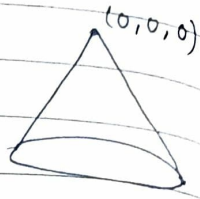
$$9(2x - y)^2 + 4(3z - y)^2 = 216$$

Date
Page

CONE

1. Find the equation of the cone whose vertex is the origin and base is the circle $x = a,\ y^2 + z^2 = b^2$.

Sol^n. Let the generator be $\frac{x}{l} = \frac{y}{m} = \frac{z}{n}$.



The generator meets the circle:

$$x = a,\ y^2 + z^2 = b^2.$$

$$\therefore \quad \frac{a}{l} = \frac{y}{m} = \frac{z}{n} \Rightarrow y = \frac{am}{l},\ z = \frac{an}{l}$$

$$\frac{a^2m^2}{l^2} + \frac{a^2n^2}{l^2} = b^2 \Rightarrow a^2\left(\frac{y^2}{x^2}\right) + a^2\left(\frac{z^2}{x^2}\right) = b^2$$

$$\Rightarrow a^2(y^2 + z^2) = b^2x^2 \quad \text{Ans.}$$

2. Find the equation of the cone whose vertex is at $(1,2,3)$ and the guiding curve is the circle $x^2 + y^2 + z^2 = 9,\ x + y + z = 1$.

Sol^n. Let the generator be $\frac{x-1}{l} = \frac{y-2}{m} = \frac{z-3}{n}$ — (1)

The generator meets the guiding curve.

$$\therefore \quad \frac{x-1}{l} = \frac{y-2}{m} = \frac{z-3}{n} = r \Rightarrow \begin{aligned} x &= lr + 1 \\ y &= mr + 2 \\ z &= nr + 3 \end{aligned}$$

$$\therefore \quad lr + 1 + mr + 2 + nr + 3 = 1$$

$$\Rightarrow (l + m + n)r = -5$$

&

$$(lr+1)^2 + (mr+2)^2 + (nr+3)^2 = 9$$

$$\Rightarrow \left\{l\cdot\left(\frac{-5}{l+m+n}\right) + 1\right\}^2 + \left\{m\left(\frac{-5}{l+m+n}\right) + 2\right\}^2$$

$$+ \left\{n\left(\frac{-5}{l+m+n}\right) + 3\right\}^2 = 9$$

$$\Rightarrow \{m + n - 4l\}^2 + \{2l - 3m + 2n\}^2$$

$$+ \{3l + 3m - 2n\}^2 = 9(l + m + n)^2 \quad \text{— (2)}$$

Eliminating l, m, n from (1) & (2) we have

$$(y - 2 + z - 3 - 4x + 4)^2 + (2x - 2 - 3y - 6 + 2z - 6)^2$$

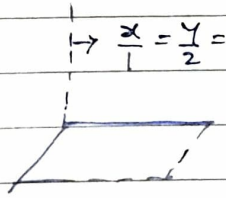
$$+ (3x - 3 + 3y - 6 - 2z - 6)^2 = 9(x + y + z - 6)^2$$

$$\Rightarrow (y+z-4x-1)^2 + (2z+2x-3y-2)^2 + (3x+3y-2z-3)^2 = 9(x+y+z-6)^2. \quad \text{Ans.}$$

3. If $\frac{x}{1} = \frac{y}{2} = \frac{z}{3}$ represents one of a set of three mutually perpendicular generators of the cone $5yz - 8zx - 3xy = 0$ find the equations of the other two.

Sol^n. The other two generators must lie on the plane which passes through origin and has $\frac{x}{1} = \frac{y}{2} = \frac{z}{3}$ as the normal.

Eq^n of such plane is $x + 2y + 3z = 0$.



Let $\frac{x}{l} = \frac{y}{m} = \frac{z}{n}$ be one of the other two generators of the cone. Then,

$$l + 2m + 3n = 0 \quad \text{---(1)}$$

and,

$$5mn - 8nl - 3lm = 0 \quad \text{---(2)}$$

Eliminating n we have,

1) $\rightarrow$ $3n = -(l+2m) \Rightarrow n = -\frac{(l+2m)}{3}$.

2) $\rightarrow$ $(5m - 8l)\left(\frac{l+2m}{-3}\right) - 3lm = 0$.

$\Rightarrow 5ml - 8l^2 + 10m^2 - 16lm + 9lm = 0$

$\Rightarrow 10m^2 - 8l^2 - 2lm = 0 \Rightarrow 5m^2 - 4l^2 - lm = 0$

$\Rightarrow 5m^2 - lm - 4l^2 = 0 \Rightarrow 5m^2 - 5lm + 4lm - 4l^2 = 0$

$\Rightarrow 5m(m-l) + 4l(m-l) = 0 \Rightarrow (5m+4l)(m-l) = 0.$

Case I $l = m$ then $n = -\frac{3m}{3} \Rightarrow n = -m$

$$\Rightarrow \frac{l}{1} = \frac{m}{1} = \frac{n}{-1}$$

Case II $5m + 4l = 0 \Rightarrow n = -\frac{1}{3}\left(-\frac{5}{4}m + 2m\right)$

$$\Rightarrow n = -\frac{1}{3}\left(\frac{3m}{4}\right) = -\frac{m}{4}.$$

$$\Rightarrow \frac{l}{5} = \frac{m}{-4} = \frac{n}{1}$$

Thus the other two generators are $\frac{x}{1} = \frac{y}{1} = \frac{z}{-1}$

and $\frac{x}{5} = \frac{y}{-4} = \frac{z}{1}$

4. Find the equations of the lines of intersection of the plane $x - 5y + 3z = 0$ (1) with the cone $7x^2 + 5y^2 - 3z^2 = 0$ (2).

Solving (1) & (2) we have

$$7x^2 + 5y^2 - 3\left(\frac{5y - x}{3}\right)^2 = 0$$

$$\Rightarrow 7x^2 + 5y^2 - \frac{1}{3}(25y^2 + x^2 - 10xy) = 0$$

$$\Rightarrow 20x^2 - 10y^2 + 10xy = 0.$$

$$\Rightarrow 20x^2 + 10xy - 10y^2 = 0 \Rightarrow 2x^2 + 2xy - xy - y^2 = 0$$

$$\Rightarrow 2x(x+y) - y(x+y) = 0 \Rightarrow (x+y)(2x - y) = 0.$$

Case I. $\frac{x}{y} = -1$ & $\frac{x}{y} = \frac{1}{2}$

from (1) for $x = -y$ we have $-6y + 3z = 0 \Rightarrow \frac{y}{2} = \frac{1}{2}z$

for $x = \frac{1}{2}y$ we have $\frac{y}{2} - 5y + 3z = 0$

$$\Rightarrow -\frac{9}{2}y + 3z = 0$$

$$\Rightarrow \frac{y}{2} = \frac{1}{3}z.$$

Thus the lines are

$$\frac{x}{-1} = \frac{y}{1} = \frac{z}{2} \quad \text{and} \quad \frac{x}{1} = \frac{y}{2} = \frac{z}{3}$$

5. Find the equation of the right circular cone whose vertex is the origin, axis is the z-axis and which passes through the point $(3, 4, -6)$.

Sol^n Let the generator be $\frac{x}{l} = \frac{y}{m} = \frac{z}{n}$.

Eq^n of z axis is $\frac{x}{0} = \frac{y}{0} = \frac{z}{1}$

If θ be the semi vertical angle then

$$\cos\theta = \frac{n}{\sqrt{l^2 + m^2 + n^2}}$$

$$\Rightarrow n^2 = \cos^2\theta\,(l^2 + m^2 + n^2).$$

Thus eq^n of the cone is

$$z^2 = \cos^2\theta\,(x^2 + y^2 + z^2)$$

it passes through $(3,4,-6)$, then,

$36(1-\cos^2\theta) = 9+16 \Rightarrow 1-\cos^2\theta = \frac{25}{36}$.

$\Rightarrow \cos^2\theta = 1 - \frac{25}{36} = \frac{9}{36}$.

$\cos^2\theta = \frac{z^2}{x^2+y^2+z^2}$

$= \frac{36}{9+16+36} = \frac{36}{61}$

∴ The equation of the cone is

$z^2 = \frac{36}{61}(x^2+y^2+z^2) \Rightarrow 36x^2 + 36y^2 = 25z^2$.

6.15 Prove that the plane $ax+by+cz=0$, $(abc \neq 0)$ cuts the cone $yz+zx+xy=0$ in perpendicular straight lines if $\frac{1}{a}+\frac{1}{b}+\frac{1}{c}=0$.

Sol. Solving (1) and (2) we have

$z = -\frac{(ax+by)}{c}$ and $(y+x)z + xy = 0$ gives

$-\frac{(y+x)(ax+by)}{c} + xy = 0 \Rightarrow ax^2 + (a+b)xy + by^2 = cxy$

$\Rightarrow ax^2 + (a+b-c)xy + by^2 = 0$. ——— (3)

Let the two st. lines be $\frac{x}{x_i} = \frac{y}{y_i} = \frac{z}{z_i}$ $i = 1, 2$

Then (3) can be written as

$a\frac{x^2}{y^2} + (a+b-c)\frac{x}{y} + b = 0$. where

$\frac{x_1x_2}{y_1y_2} = \frac{b}{a}$ & $\frac{x_1}{y_1} + \frac{x_2}{y_2} = \frac{x_1y_2+y_1x_2}{y_1y_2} = -\frac{(a+b-c)}{a}$

——— (4)

Angle between the lines

$z_1z_2 = +\frac{(ax_1+by_1)}{c}\frac{(ax_2+by_2)}{c}$

$= \frac{1}{c^2}\left[a^2x_1x_2 + ab(x_1y_2+y_1x_2) + b^2y_1y_2\right]$

$$\frac{z_1 z_2}{y_1 y_2} = \frac{1}{c^2}\left[a^2 \frac{x_1 x_2}{y_1 y_2} + ab\left(\frac{x_1 y_2 + y_1 x_2}{y_1 y_2}\right) + b^2\right] \quad \text{---(5)}$$

Angles between the lines is

$$\cos\theta = \frac{x_1 x_2 + y_1 y_2 + z_1 z_2}{\sqrt{x_1^2 + y_1^2 + z_1^2}\sqrt{x_2^2 + y_2^2 + z_2^2}}$$

If $\theta = 90°$ then $x_1 x_2 + y_1 y_2 + z_1 z_2 = 0$

or, $\frac{x_1 x_2}{y_1 y_2} + 1 + \frac{z_1 z_2}{y_1 y_2} = 0$

from (4) and (5) we have

$$1 + \frac{b}{a} + \frac{1}{c^2}\left[a^2\left(\frac{b}{a}\right) + ab\left\{-\frac{(a+b-c)}{a}\right\} + b^2\right] = 0$$

$$\Rightarrow 1 + \frac{b}{a} + \frac{1}{c^2}\left[ab + b(c-a-b) + b^2\right] = 0$$

$$\Rightarrow 1 + \frac{b}{a} + \frac{1}{c^2}\left[ab + bc - ab - b^2 + b^2\right] = 0$$

$$\Rightarrow 1 + \frac{b}{a} + \frac{b}{c} = 0 \Rightarrow \frac{1}{b} + \frac{1}{a} + \frac{1}{c} = 0$$

$$\Rightarrow \frac{1}{a} + \frac{1}{b} + \frac{1}{c} = 0 \quad \text{proved.}$$

7. Show that the eqⁿ of the cone whose generators pass through the point (α, β, γ) and have their direction cosines satisfying the relation $al^2 + bm^2 + cn^2 = 0$ is $a(x-\alpha)^2 + b(y-\beta)^2 + c(z-\gamma)^2 = 0$. ---(1)

Sol^n: Eqⁿ of the generators is

$$\frac{x-\alpha}{l_1} = \frac{y-\beta}{m_1} = \frac{z-\gamma}{n_1} \quad \text{---(2)}$$

where l_1, m_1, n_1 may be thought to be d.c's.

Since l_1, m_1, n_1 satisfies the relation (1)

$\therefore a l_1^2 + b m_1^2 + c n_1^2 = 0$

Thus locus of (2) is $a(x-\alpha)^2 + b(y-\beta)^2 + c(z-\gamma)^2 = 0$.

Date
Page

8.8. A variable plane through the x-axis and a variable plane through the y-axis are inclined at a constant angle α. Prove that** $z^2(x^2+y^2+z^2) = x^2y^2\tan^2\alpha$.
** their line of intersection generates the cone

Sol^n. Eq^n of plane through x-axis is $By + C_1 z = 0$
" " " " y-axis " $Ax + C_2 z = 0$

Then, $\cos\alpha = \dfrac{C_1 C_2}{\sqrt{B^2+C_1^2}\sqrt{A^2+C_2^2}}$

$\Rightarrow \sec^2\alpha = \dfrac{(B^2+C_1^2)(A^2+C_2^2)}{C_1^2 C_2^2} = \left\{\left(\dfrac{B}{C_1}\right)^2+1\right\}\left\{1+\left(\dfrac{A}{C_2}\right)^2\right\}$.

$\Rightarrow \tan^2\alpha = \left(1+\dfrac{B^2}{C_1^2}\right)\left(1+\dfrac{A^2}{C_2^2}\right) - 1$ ——— (1).

Eq^n of the line of intersection is :
If l, m, n are the d.r. of the line then

$0\cdot l + Bm + C_1 n = 0$
$Al + 0\cdot m + C_2 n = 0$

$\Rightarrow \dfrac{l}{BC_2} = \dfrac{m}{C_1 A} = \dfrac{n}{-AB}$

If $z=0$ then $y=0$ & $x=0$ then the st. line is

$$\frac{x}{BC_2} = \frac{y}{C_1 A} = \frac{z}{-AB} \quad \text{——— (2)}$$

$\therefore$ locus of (2) is the cone.

$\dfrac{x}{z} = -\dfrac{C_2}{A} \Rightarrow \dfrac{z}{x} = -\dfrac{A}{C_2}$

$\dfrac{y}{z} = -\dfrac{C_1}{B} \Rightarrow \dfrac{z}{y} = -\dfrac{B}{C_1}$

from (1). $\tan^2\alpha = \left(1+\dfrac{z^2}{y^2}\right)\left(1+\dfrac{z^2}{x^2}\right) - 1$

$\Rightarrow \tan^2\alpha = 1 + \dfrac{z^4}{x^2y^2} + \dfrac{z^2}{y^2} + \dfrac{z^2}{x^2} - 1$

$\Rightarrow \tan^2\alpha\,(x^2y^2) = z^2(x^2+y^2+z^2)$

Quadric Surfaces.

(Central conicoids).

Ellipsoids : $\frac{x^2}{a^2} + \frac{y^2}{b^2} + \frac{z^2}{c^2} = 1$. $(a, b, c \neq 0)$. —— (1)

Note : (1) The surface is symmetrical w.r.t. origin because : (α, β, γ) and $(-\alpha, -\beta, -\gamma)$ both lies on (1)

(2) The surface is symmetrical w.r.t xy, yz and zx plane because the points $\{(\alpha, \beta, \gamma), (\alpha, \beta, -\gamma)\}$, $\{(\alpha, \beta, \gamma), (-\alpha, \beta, \gamma)\}$ and $\{(\alpha, \beta, \gamma), (\alpha, -\beta, \gamma)\}$ all lie on (1).

The planes $x=0$, $y=0$ and $z=0$ are the principal planes and the lines $x=0=y$, $x=0=z$, $y=0=z$ are the principal axes.

(3) The surface is symmetrical w.r.t. the principal axes.

(4) The ellipsoid is generated by a variable ellipse because the sections of (1) by the planes $x=k_1$, $y=k_2$ & $z=k_3$ are all ellipses.

Hyperboloids : $\frac{x^2}{a^2} + \frac{y^2}{b^2} - \frac{z^2}{c^2} = 1$ (of one sheet)

$(a, b, c \neq 0)$ $\frac{x^2}{a^2} - \frac{y^2}{b^2} + \frac{z^2}{c^2} = 1$ "

$-\frac{x^2}{a^2} + \frac{y^2}{b^2} + \frac{z^2}{c^2} = 1$ "

Note : (1) Symmetric w.r.t. origin, co-ord. planes and the co-ord. axes.

(2). The surface meets the x axis at $(-a, 0, 0)$ & $(0, 0, a)$., the y axis at $(0, b, 0)$ & $(0, -b, 0)$ but does not meet the z-axis in real points.

(3) The sections of the surface by the planes $z=k_1$, $x=k_2$ & $y=k_3$ are ellipse & hyperbola respectively.

$$\frac{x^2}{a^2} - \frac{y^2}{b^2} - \frac{z^2}{c^2} = 1 \quad \text{(of two sheets)}$$

$$-\frac{x^2}{a^2} - \frac{y^2}{b^2} + \frac{z^2}{c^2} = 1 \quad \text{"}$$

& $$-\frac{x^2}{a^2} + \frac{y^2}{b^2} - \frac{z^2}{c^2} = 1 \quad \text{"}$$

$(a, b, c \neq 0)$.

Note : (1) Symmetric w.r.t. origin, co-ord planes and co-ord. axes.

(2) The surface meets the x-axis at $(-a, 0, 0)$ & $(a, 0, 0)$ and does not meet the y & z axes at real pts.

(3) the section by the plane $z = K$ is a hyperbola, by the plane $y = K$ is a hyperbola & by the plane $x = K$ is an ellipse.

Paraboloids (not central conicoid) :

$$\frac{x^2}{a^2} + \frac{y^2}{b^2} = \frac{2z}{c} \quad \text{(elliptic paraboloid)} \quad \text{—— (1)}$$

& $$\frac{x^2}{a^2} - \frac{y^2}{b^2} = \frac{2z}{c} \quad \text{(hyperbolic paraboloid)} \quad \text{—— (2)}$$

Note : (1) The section of the surface (1) by the plane $z = K$ is an ellipse and the surface (2) is a hyperbola. The sections by the planes $x = K$ & $y = K$ are parabolas.

(2) The surfaces are symmetric w.r.t. the yz and xz planes.

(3) The surface (1) always lies on the positive side of the plane $z = 0$ because z is always +ve.

General Equation of central conicoids : $ax^2+by^2+cz^2=1$

" " " paraboloids : $ax^2+by^2=2cz$

Problems:

1. Find the nature of the conic quadratic surface
$2x^2+5y^2+3z^2-4x+20y-6z=5$

Sol^n.
$$2(x^2-2x)+5(y^2+4y)+3(z^2-2z)=5$$
$$\Rightarrow 2(x-1)^2+5(y+2)^2+3(z-1)^2=5+2+20+3$$
$$=30$$

Shifting the origin to $(1,-2,1)$ we have
$X=x-1$, $Y=y+2$, $Z=z-1$ and

$$2X^2+5Y^2+3Z^2=30$$
$$\Rightarrow \frac{X^2}{15}+\frac{Y^2}{6}+\frac{Z^2}{10}=1$$ represents an ellipsoid.

2. Find the nature of the conicoid
$3x^2-2y^2-12x-12y-6z=0$.

Sol^n.
$$3(x^2-4x)-2(y^2+6y)-6z=0$$
$$\Rightarrow 3(x-2)^2-2(y+3)^2=6z+12\mp 18.$$
$$\Rightarrow 3(x-2)^2-2(y+3)^2=6(z-1).$$

$X=x-2$ $Y=y+3$ $Z=z-1$ gives
$$3X^2-2Y^2=6Z.$$
$$\Rightarrow \frac{X^2}{2}-\frac{Y^2}{3}=\frac{2Z}{2}$$ represents a hyperbolic paraboloid.

GENERATING STRAIGHT LINES.

Date
Page

Ruled Surfaces & Generating straight lines : If through every point on a surface a straight line can be drawn such that it always lies wholly on the surface then such a line is a generator and the surface generated is a ruled surface.

Generating lines of a hyperboloid of one sheet :

$$\frac{x^2}{a^2} + \frac{y^2}{b^2} - \frac{z^2}{c^2} = 1 \qquad \text{---(1)}$$

$$\Rightarrow \frac{x^2}{a^2} - \frac{z^2}{c^2} = 1 - \frac{y^2}{b^2} \Rightarrow \left(\frac{x}{a} - \frac{z}{c}\right)\left(\frac{x}{a} + \frac{z}{c}\right) = \left(1 - \frac{y}{b}\right)\left(1 + \frac{y}{b}\right)$$

Consider the straight lines

$$\frac{x}{a} - \frac{z}{c} = \lambda\left(1 - \frac{y}{b}\right), \quad \frac{x}{a} + \frac{z}{c} = \frac{1}{\lambda}\left(1 + \frac{y}{b}\right) \qquad \text{---(2)}$$

and $$\frac{x}{a} - \frac{z}{c} = \mu\left(1 + \frac{y}{b}\right), \quad \frac{x}{a} + \frac{z}{c} = \frac{1}{\mu}\left(1 - \frac{y}{b}\right) \qquad \text{---(3)}$$

Thus (2) & (3) are two sets of generators of (1).

Generating lines of a hyperbolic paraboloid :

$$\frac{x^2}{a^2} - \frac{y^2}{b^2} = 2z \qquad \text{---(1)}$$

$$\Rightarrow \left(\frac{x}{a} - \frac{y}{b}\right)\left(\frac{x}{a} + \frac{y}{b}\right) = 2z$$

Consider the st. lines

$$\frac{x}{a} - \frac{y}{b} = 2\lambda z, \quad \frac{x}{a} + \frac{y}{b} = \frac{1}{\lambda}$$

and $$\frac{x}{a} - \frac{y}{b} = 2\mu, \quad \frac{x}{a} + \frac{y}{b} = \mu z.$$

Problems.

1. Find the equations of the generating lines of the hyperboloid $\frac{x^2}{4} + \frac{y^2}{9} - \frac{z^2}{16} = 1$ which passes through the point $(2, 3, -4)$.

Sol^n. Let the eq^n of the generator be

$$\left(\frac{x}{2} - \frac{z}{4}\right) = \lambda\left(1 - \frac{y}{3}\right) \quad ; \quad \frac{x}{2} + \frac{z}{4} = \frac{1}{\lambda}\left(1 + \frac{y}{3}\right).$$

It passes through $(2, 3, -4)$ then

$$\left(\frac{2}{2} + \frac{4}{4}\right) = \lambda\left(1 - \frac{3}{3}\right) \Rightarrow 2 = \lambda(1-1) \Rightarrow 0 = \frac{1}{\lambda}$$

& $\left(\frac{2}{2} - \frac{4}{4}\right) = \frac{1}{\lambda}\left(1 + \frac{3}{3}\right) \Rightarrow 0 = \frac{1}{\lambda}$ not permissible

Then $\frac{x}{2} - \frac{z}{4} = \frac{1}{\lambda}\left(1 - \frac{y}{3}\right)$; $\frac{x}{2} + \frac{z}{4} = \lambda\left(1 + \frac{y}{3}\right)$

gives $\lambda = 0$.

Thus the generator is

$1 - \frac{y}{3} = 0$; $\frac{x}{2} + \frac{z}{4} = 0$.

$\Rightarrow y = 3$, $2x + z = 0$.

Again $\frac{x}{2} - \frac{z}{4} = \mu\left(1 + \frac{y}{3}\right)$; $\frac{x}{2} + \frac{z}{4} = \frac{1}{\mu}\left(1 - \frac{y}{3}\right)$

Since $(2, 3, -4)$ lies on the generating lines

then, $\frac{2}{2} + \frac{4}{4} = \mu\left(1 + \frac{3}{3}\right) \Rightarrow 2 = 2\mu \Rightarrow \mu = 1$

and $\frac{2}{2} - \frac{4}{4} = \frac{1}{\mu}\left(1 - \frac{3}{3}\right) \Rightarrow$.

$\therefore$ $2x - z = 4\left(1 + \frac{y}{3}\right)$; $2x + z = 4\left(1 - \frac{y}{3}\right)$.

Tangents and Normals

3-D

Equation of tangent line & tangent Plane to a central conicoid.

Let the conicoid be $ax^2 + by^2 + cz^2 = 1$ —(1) & the point (x_1, y_1, z_1)

Eq^n of st. line through (x_1, y_1, z_1) having d.c. l, m, n is

$$\frac{x - x_1}{l} = \frac{y - y_1}{m} = \frac{z - z_1}{n} \quad \text{—(2)}$$

any pt. on it is $(lr + x_1, mr + y_1, nr + z_1)$.

(2) will int. This pt. will lie on (1) if

$$a(x_1 + lr)^2 + b(y_1 + mr)^2 + c(z_1 + nr)^2 = 1$$

or, $(al^2 + bm^2 + cn^2)r^2 + 2r(alx_1 + bmy_1 + cnz_1)$

$$+ ax_1^2 + by_1^2 + cz_1^2 - 1 = 0. \quad \text{—(3)}$$

Since (3) is a quadratic in r so (2) intersects (1) at 2 pts. for $r = r_1$ and r_2.

If (2) touches (1) at (x_1, y_1, z_1) then distance of the two pts. from (x_1, y_1, z_1) is zero. i.e $r_1 = r_2 = 0$.

Roots of (3) will be equal to zero if

$$alx_1 + bmy_1 + cnz_1 = 0 \quad \text{—(4)}$$

& $ax_1^2 + by_1^2 + cz_1^2 - 1 = 0$. —(5).

Thus (2) along with (4) & (5) is a tangent line to (1).

Tangent Plane : Locus of (2) along with cond.^{ns} (4) & (5) is a tangent plane. we eliminate l, m, n.

$$ax_1(x - x_1) + by_1(y - y_1) + cz_1(z - z_1) = 0$$

$$\Rightarrow axx_1 + byy_1 + czz_1 - (ax_1^2 + by_1^2 + cz_1^2) = 0$$

$$\Rightarrow axx_1 + byy_1 + czz_1 = 1.$$

2. Eqn of tangent plane to a cone.

Let the eqn of cone having vertex at origin be

$$ax^2 + by^2 + cz^2 + 2fyz + 2gzx + 2hxy = 0. \quad \text{—(1)}.$$

Let the tangent be at (α, β, γ). and let (1) be represented as $F(x, y, z) = 0$. —(2)

Eqn of st. line through (α, β, γ) & having d.c. l, m, n is

$$\frac{x-\alpha}{l} = \frac{y-\beta}{m} = \frac{z-\gamma}{n} \quad \text{—(3)}.$$

any pt. on (3) is $(lr+\alpha, mr+\beta, nr+\gamma)$

This will lie on (1) if

$$a(\alpha+lr)^2 + b(\beta+mr)^2 + c(\gamma+nr)^2 + 2f(mr+\beta)(nr+\gamma) + 2g(lr+\alpha)(nr+\gamma) + 2h(lr+\alpha)(mr+\beta) = 0. \quad \text{—(4)}$$

The coeff. of

r^2 : $al^2 + bm^2 + cn^2 + 2fmn + 2gln + 2hlm = F(l, m, n)$.

$2r$: $a\alpha l + b\beta m + c\gamma n + fm\gamma + f\beta n + gl\gamma + g\alpha n + hl\beta + hm\alpha$

$= l(a\alpha + g\gamma + h\beta) + m(b\beta + f\gamma + h\alpha) + n(c\gamma + f\beta + g\alpha)$

$\therefore$ (4) can be written as

$$F(l, m, n)\, r^2 + 2r\{l(a\alpha + h\beta + g\gamma) + m(h\alpha + b\beta + f\gamma) + n(g\alpha + f\beta + c\gamma)\} + F(\alpha, \beta, \gamma) = 0. \quad \text{—(5)}$$

For (3) to be a tangent to (1) at (α, β, γ), the two roots of (4) must vanish. i.e. from (5)

$$l(a\alpha + h\beta + g\gamma) + m(h\alpha + b\beta + f\gamma) + n(g\alpha + f\beta + c\gamma) = 0 \quad \text{—(6)}$$

$$\Rightarrow (y+z-4x-1)^2 + (2z+2x-3y-2)^2 + (3x+3y-2z-3)^2 = 9(x+y+z-6)^2. \quad \text{Ans.}$$

To obtain the tangent plane we eliminate l, m, n from (3) & (6) to get

$$(x-\alpha)(a\alpha+h\beta+g\gamma) + (y-\beta)(h\alpha+b\beta+f\gamma) + (z-\gamma)(g\alpha+f\beta+c\gamma) = 0.$$

$$\Rightarrow x(a\alpha+h\beta+g\gamma) + y(h\alpha+b\beta+f\gamma) + z(g\alpha+f\beta+c\gamma) = 0.$$

Normal to a conicoid.

Let the conicoid be $ax^2+by^2+cz^2=1$. ——(1)

Eqn of tangent ~~to (1)~~ plane to (1) at (α, β, γ) is

$$ax\alpha + by\beta + cz\gamma = 1. \quad \text{——(2).}$$

Eqn of normal to (1) is the st. line $\perp$ to (2) & passing through (α, β, γ).

$$\frac{x-\alpha}{a\alpha} = \frac{y-\beta}{b\beta} = \frac{z-\gamma}{c\gamma}$$

Enveloping Cone.

The locus of st. lines tangent to a given surface and drawn from a given point is called enveloping cone or tangent cone to that surface.

Eq^n Enveloping cone of the conicoid $ax^2+by^2+cz^2=1$, from (α, β, γ).

Sol^n. Eqn of st. line through (α, β, γ) & d.c. l, m, n is

$$\frac{x-\alpha}{l} = \frac{y-\beta}{m} = \frac{z-\gamma}{n} \quad \text{——(1).}$$

This line will intersect $ax^2+by^2+cz^2=1$ at $(lr+\alpha, mr+\beta, nr+\gamma)$ ——(2) for two values of r if

$$\frac{x+3}{12} = \frac{y-5}{2} = \frac{z-2}{2}$$

$$a(lr+\alpha)^2 + b(mr+\beta)^2 + c(nr+\gamma)^2 = 1$$

$$\text{i.e., } r^2(al^2+bm^2+cn^2) + 2r(al\alpha+bm\beta+cn\gamma) + a\alpha^2+b\beta^2+c\gamma^2-1=0 \quad \text{——(3)}$$

The st line (1) will be a tangent to (2) if the 2 values of r are equal.

$$\text{i.e } (al\alpha+bm\beta+cn\gamma)^2 = (al^2+bm^2+cn^2)(a\alpha^2+b\beta^2+c\gamma^2-1) \quad \text{——(4)}$$

To obtain the locus of (1), eliminate l, m, n between (3) & (4)

$$\text{i.e } \{a\alpha(x-\alpha)+b\beta(y-\beta)+c\gamma(z-\gamma)\}^2$$

$$= \{a(x-\alpha)^2+b(y-\beta)^2+c(z-\gamma)^2\}(a\alpha^2+b\beta^2+c\gamma^2-1).$$

This is the equation of the enveloping cone with (α, β, γ) as its vertex.

14. (a). Eq^n of tangent plane to $\frac{x^2}{a^2} + \frac{y^2}{b^2} + \frac{z^2}{c^2} = 1$ at (α, β, γ) is

$$\frac{x\alpha}{a^2} + \frac{y\beta}{b^2} + \frac{z\gamma}{c^2} = 1 \quad \text{—— (1)}$$

Eq^n of given plane : $lx + my + nz = p$ —— (2)

(1) & (2) are identical if

$$\frac{\alpha/a^2}{l} = \frac{\beta/b^2}{m} = \frac{\gamma/c^2}{n} = \frac{1}{p}.$$

$$\Rightarrow \alpha = \frac{a^2 l}{p}, \quad \beta = \frac{b^2 m}{p}, \quad \gamma = \frac{c^2 n}{p}.$$

Since (α, β, γ) lies on the ellipsoid

$$\therefore \frac{a^4 l^2}{p^2 a^2} + \frac{b^4 m^2}{p^2 b^2} + \frac{c^4 n^2}{c^2 p^2} = 1$$

$$\Rightarrow a^2 l^2 + b^2 m^2 + c^2 n^2 = p^2$$

17. The required cone is the enveloping cone to the sphere $x^2 + y^2 + z^2 = 25$. with $(6, 0, 0)$ as vertex.

$a = 1/25 \quad b = 1/25 \quad c = 1/25.$

$$\left[a\alpha(x-\alpha) + b\beta(y-\beta) + c\gamma(z-\gamma)\right]^2$$

$$= \left[a(x-\alpha)^2 + b(y-\beta)^2 + c(z-\gamma)^2\right](a\alpha^2 + b\beta^2 + c\gamma^2 - 1).$$

$$\Rightarrow \left\{\frac{1}{25} \times 6(x-6)\right\}^2 = \left[\frac{1}{25}(x-6)^2 + \frac{y^2}{25} + \frac{z^2}{25}\right]\left(\frac{36}{25} - 1\right)$$

$$\Rightarrow \frac{36}{25^2}(x-6)^2 = \frac{11}{25^2}(x-6)^2 + \frac{11y^2}{25^2} + \frac{11z^2}{25^2}$$

$$\Rightarrow 25(x-6)^2 - 11y^2 - 11z^2 = 0.$$

$$\Rightarrow 25x^2 - 11y^2 - 11z^2 - 300x + 900 = 0$$

8(a). $(x-4)+\lambda(3y+4z)=0$ is the req. tangent plane. —(1)

Let (α,β,γ) be a pt on $x^2+3y^2-6z^2-4=0$. —(2)

Eq^n of tangent plane at (α,β,γ) at (2) is $x\alpha+3\beta y-6\gamma z-4=0$ —(3).

(1) & (3) are identical if

$$\frac{\alpha}{1}=\frac{3\beta}{3\lambda}=\frac{-6\gamma}{4\lambda}=\frac{-4}{-4}.$$

$\Rightarrow \alpha=1$ $\lambda=\beta$ $-6\gamma=4\lambda \Rightarrow \gamma=-\frac{2}{3}\lambda$.

from (2) $\alpha^2+3\beta^2-6\gamma^2-4=0$

$\Rightarrow 1^2+3\lambda^2-6\cdot\frac{4}{9}\lambda^2=4.$

$\Rightarrow (27-24)\lambda^2=27 \Rightarrow 3\lambda^2=27 \Rightarrow \lambda=\pm 3$

Eq^n of planes are for $\lambda=3$ $x+9y+12z-4=0$

$\lambda=-3$ $x-9y-12z-4=0$.